第二节：传感器与成像几何基础

杨聪

目录 CONTENTS

计算机视觉的研究内容与体系

➢ 什么是计算机视觉（Computer Vision）？

计算机视觉是以图像（视频）为输入，以对环境的表达（representation）和理解为目标，研究图像信息组织、物体和场景识别、进而对事件给予解释的学科。

➢ 视觉分为三个阶段

- 特征提取和区域分割
  - 基于轮廓,纹理,颜色…
- 建模与模式表达
  - 基于各种物体的抽象化模型
- 描述和理解
  - 基于景物的结构知识

底层处理　少

中层处理　知识

高层处理　多

胡占义. 计算机视觉简介：历史、现状和发展趋势. https://mp.weixin.qq.com/s/If1GdkPuxHXLU4pr0UDaHA

研究内容与体系

01

输入设备

输入设备(input device)的研制,包括成像设备和数字化设备。

02

预处理

用图像处理技术和算法，对输入的原始图像预处理，抽取场景基本特征。

03

建三维描述

恢复场景相关的2.5维信息，以此恢复物体的完整三维图，建三维描述。

04

物体识别

根据预先存贮的模型知识以及有关特征,识别图像中物体,并确定类别。

05

拓扑关系图

建立各个图像中物体的拓扑关系图,给出图像所反映景物的结构描述。

06

体系结构

并行结构、分层结构、信息流结构、拓扑结构以及从设计到实现的途径。

计算机视觉

- 视觉基础
  - ◆ 摄像机
  - ◆ 摄像机的几何模型
  - ◆ 摄像机的几何标定
  - ◆ 辐射学-光亮度度量
  - ◆ 光源、阴影与影调
  - ◆ 颜色
- 低层视觉
  - ● 线性滤波
  - ● 边缘检测
  - ● 纹理
- 中层处理
  - ■ 图像的特征提取
  - ■ 目标检测
  - ■ 图像分割
  - ■ 目标跟踪
- 高层处理
  - ➢ 图像分类与识别
  - ➢ 图像检索
  - ➢ 图像captioning
  - ➢ 图像理解

目录

CONTENTS

视觉传感器

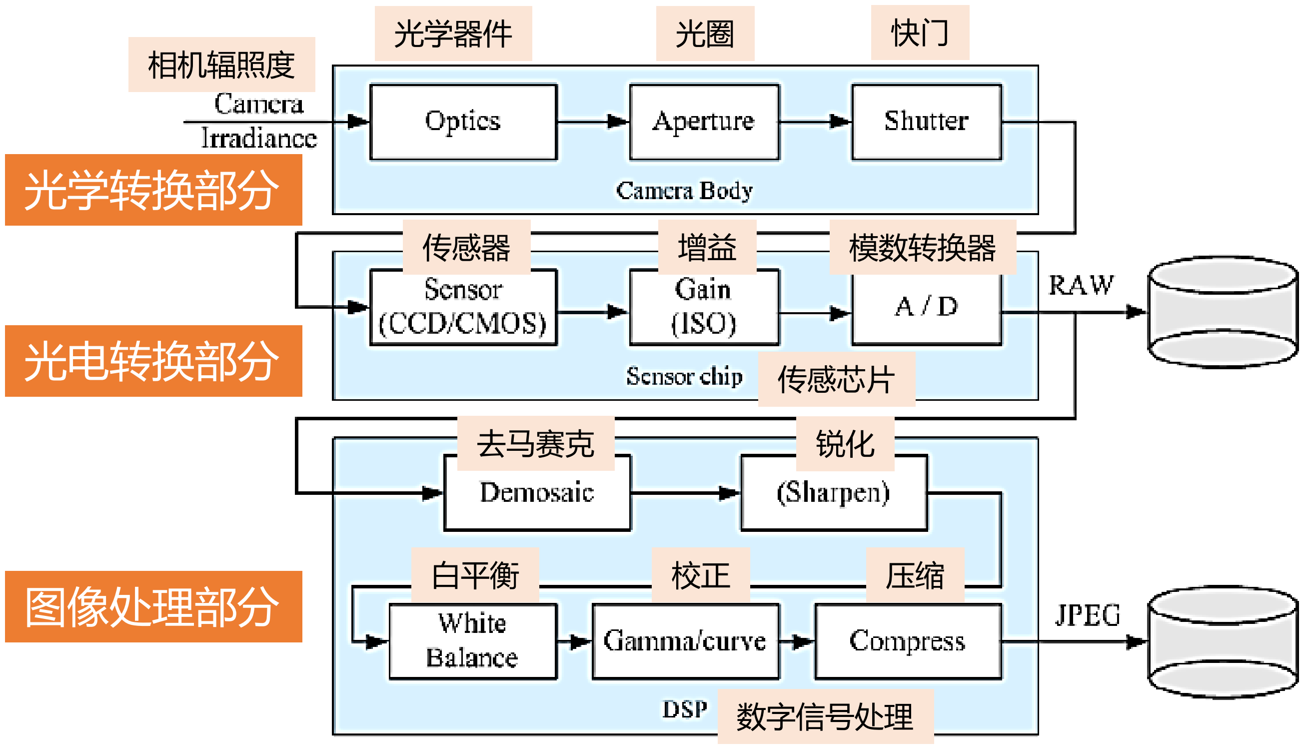
定义

利用光学元件和成像装置获取外部环境图像信息的仪器，通常用图像分辨率来描述视觉传感器的性能。视觉传感器的精度不仅与分辨率有关，而且同被测物体的检测距离相关。被测物体距离越远，其绝对的位置精度越差。

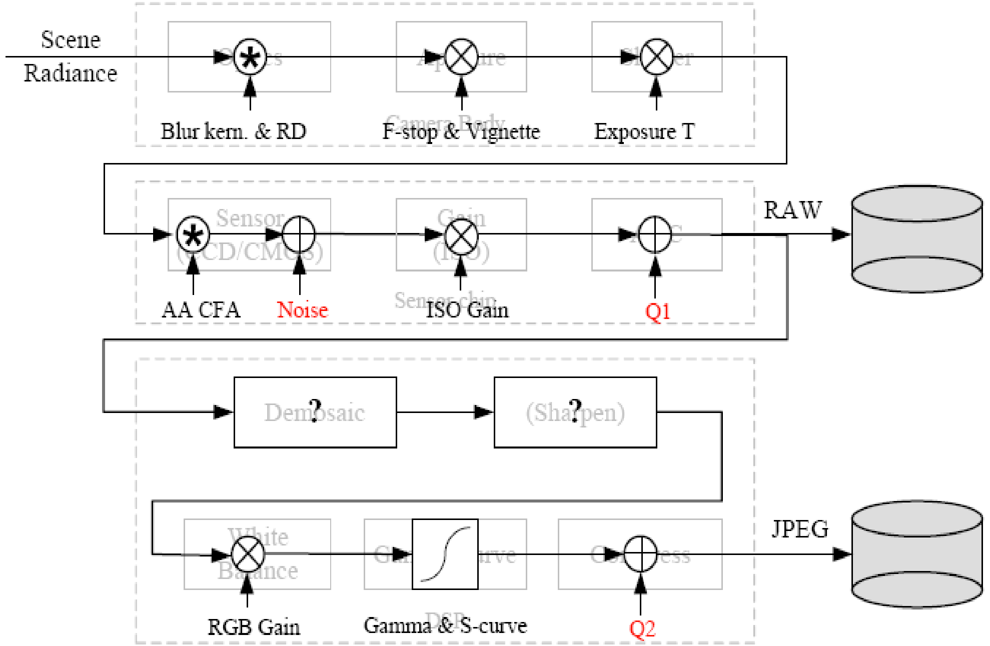
组成

视觉传感器是整个机器视觉系统信息的直接来源，主要由一个或者两个图形传感器组成，有时还要配以光投射器及其他辅助设备。视觉传感器的主要功能是获取足够的机器视觉系统要处理的最原始图像。图像传感器可以使用激光扫描器、线阵和面阵CCD摄像机或者TV摄像机，也可以是最新出现的数字摄像机等。

数码相机

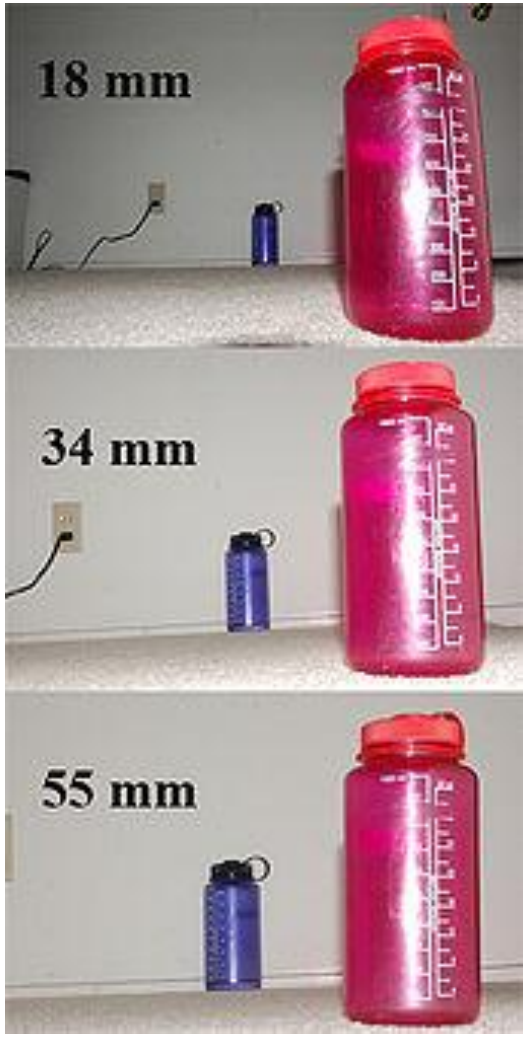


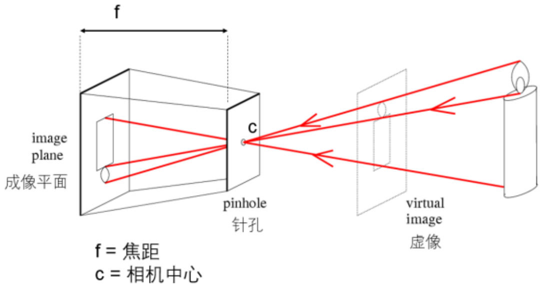
数码相机构造



数码相机模型

焦距 focal length





光圈Aperture

- 对于已制造好的镜头，镜头的直径不能改变，但可以在镜头内部加入多边形或者圆形且面积可变的孔状光栅来达到控制镜头通光量，这个装置就叫做光圈。

- 光圈的大小值直接影响到进光量和景深：
  - 光圈F值越小表示光圈越大。
  - 相同的ISO与快门条件下，光圈越大画面也就会越亮。
  - 光圈与景深的关系：
    ① F2.8大光圈，常用于营造背景模糊的浅景深；
    ② 使用F16小圈，在拍摄主体与背景时都会比较清晰。

数码相机-光学转换部分

快门shutter：快门是摄像器材中用来控制光线照射感光元件时间的装置。

- 快门速度代表着曝光时间的长短，通常在光线充足的条件下，所需的曝光时间越短；光线不足的状况下，所需的曝光时间越长。
- 通常长时间曝光需要搭配脚架来稳定相机，让影像不会产生晃动的残影。

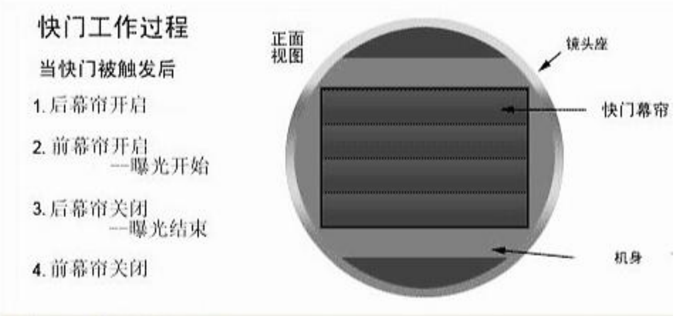
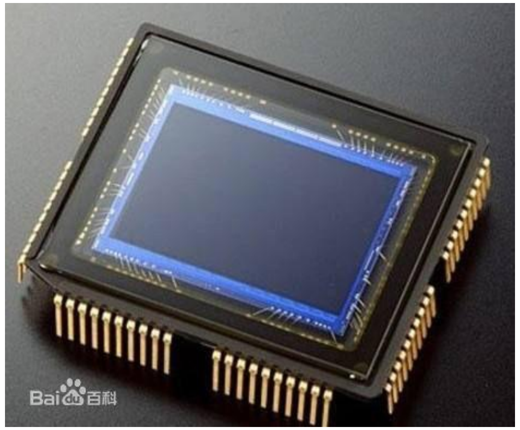


Fig 1.6.1: 快门工作过程

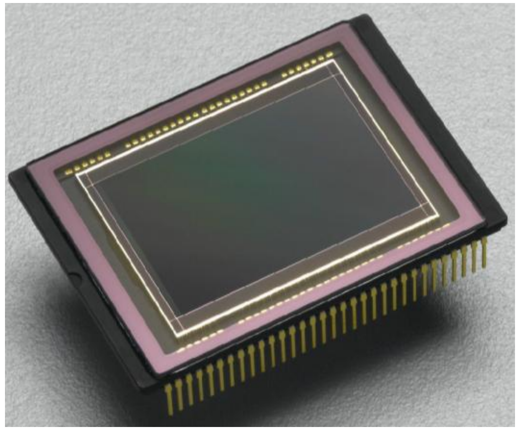
曝光时间短，画面是冻结静止的状态 (ISO 100, F4, 0.6s)

曝光时间长，记录下流动的车灯轨迹 (ISO 100, F16, 8s)

- 光电传感器
  - CCD：电荷耦合元件
  - CMOS：互补金属氧化物半导体



CCD

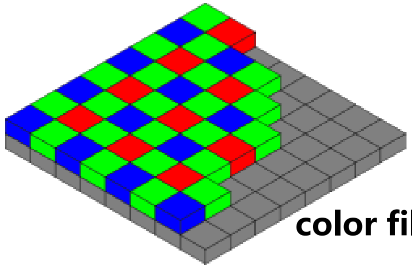


CMOS

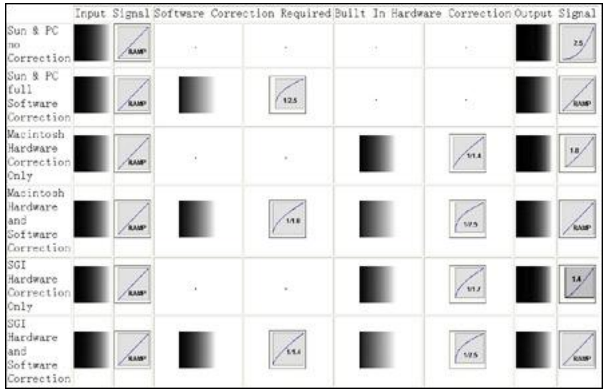
- AGC: 自动增益控制
  - ISO 感光度设置：100，200，400，...
  - ISO 数值越高接受的光量也就越多，在相同的光圈与快门条件下，画面会随着感亮度增加而明亮。相对的画面的噪声(颗粒感)也会增加。

ISO越高杂质就越多，影响了图像色彩。

[新手教学]光圈、快门、ISO你了解多少. http://www.ssxf.net/student/classroom/20120308503.htm



color filter array：Bayer mode



- 去马赛克Demosaicing
  - 从CFA输出的空间欠采样颜色通道重建全色图像（即全套三色）。
- 白平衡White Balance
  - 也叫Color Balance, 是指对三原色的全局调整，使得能够适应不同光照条件。
- Gamma 校正
  - 校正补偿不同输出设备存在的颜色显示差异，从而使图像在不同的监视器上呈现出相同的效果。

➢ 压缩Compress （JPEG，JPEG2000，MPEG7）

主要步骤：

- 将图像由RGB空间映射到YCrCb空间；
- 将各颜色分量图像分为*N*个小块，计算每一块中所有象素各颜色分量的平均值，并以此作为该块的代表颜色；
- 将各块平均值数据做DCT(小波)变换；
- 通过之字形扫描和量化，取出3组颜色DCT(小波)变换后的低频分量，共同构成该图像的颜色布局描述符。

目录

CONTENTS

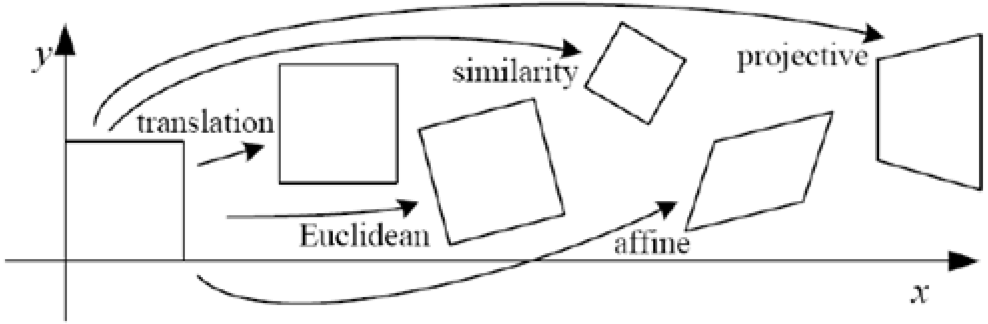
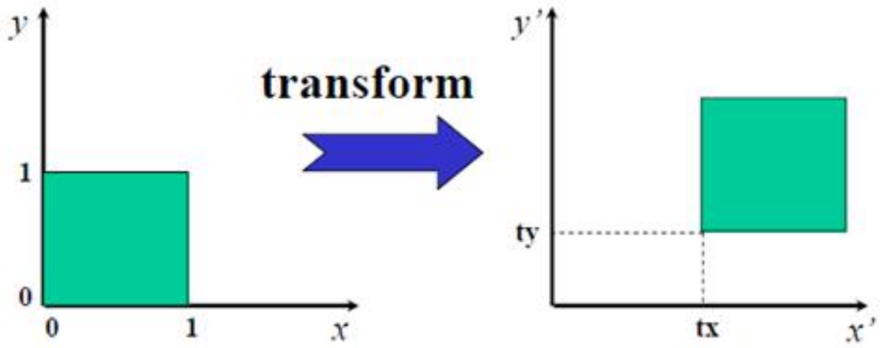


FIGURE 1. Basic set of 2D planar transformations



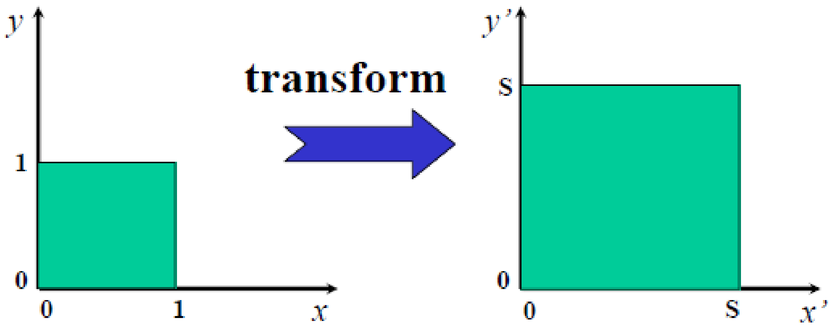
1. 平移

$$x' = x + t_x$$
$$y' = y + t_y$$

equations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

matrix form



2. 尺度变换

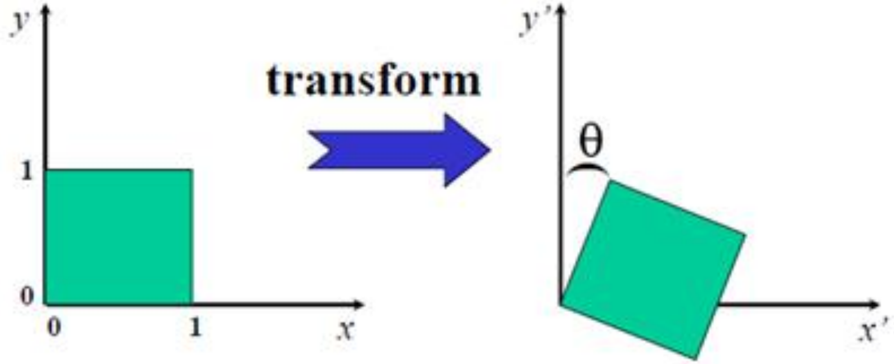
$$x' = s\, x_i$$
$$y' = s\, y_i$$

equations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

matrix form

3. 旋转



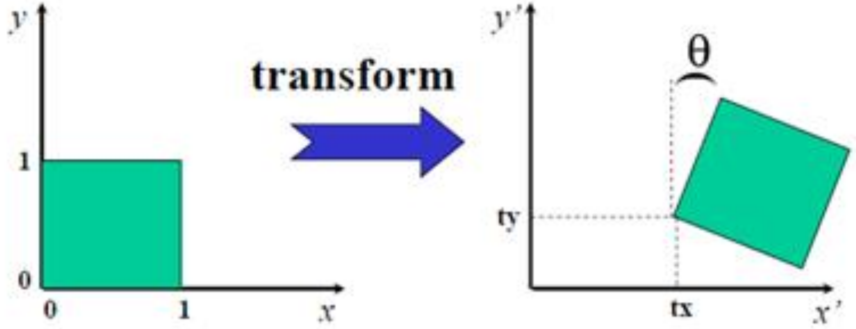
$$x' = x_i\cos\theta - y_i\sin\theta$$
$$y' = x_i\sin\theta + y_i\cos\theta$$

equations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

matrix form

4. 刚体运动



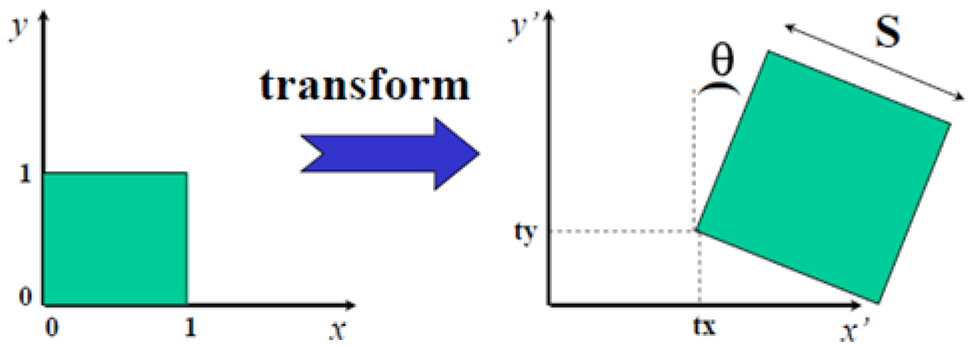
$$x' = x_i\cos\theta - y_i\sin\theta + t_x$$
$$y' = x_i\sin\theta + y_i\cos\theta + t_y$$

equations

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

matrix form

5. 刚体+尺度变换



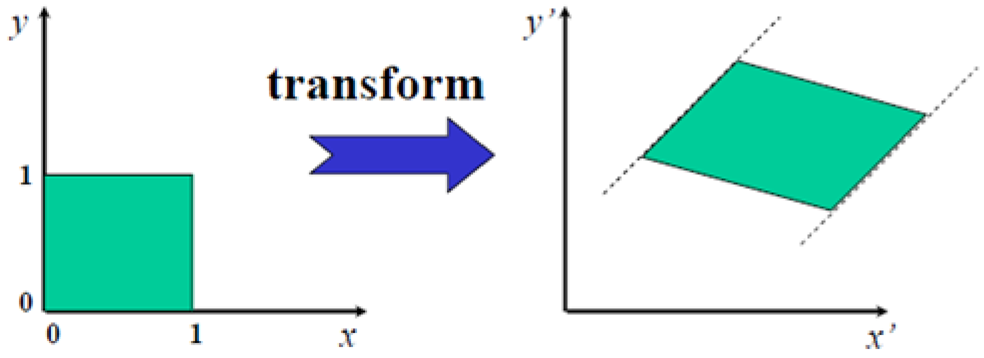
$$p' = sRp + t$$

equations

$$\begin{bmatrix} p' \\ 1 \end{bmatrix} = \begin{bmatrix} sR & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix}$$

matrix form

6. 仿射变换



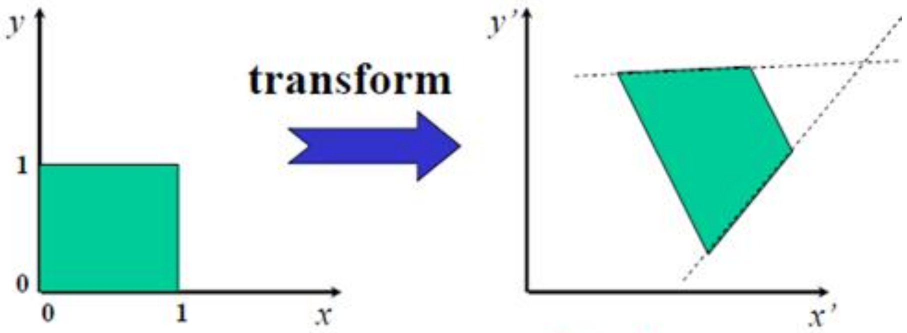
$$p' = Ap + b$$

equations

$$\begin{bmatrix} p' \\ 1 \end{bmatrix} = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix}$$

matrix form

7. 投影变换



Note!

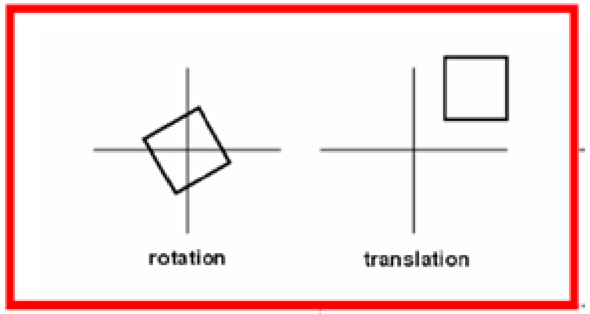
$$p' = \frac{Ap+b}{c^T p+1}$$

equations

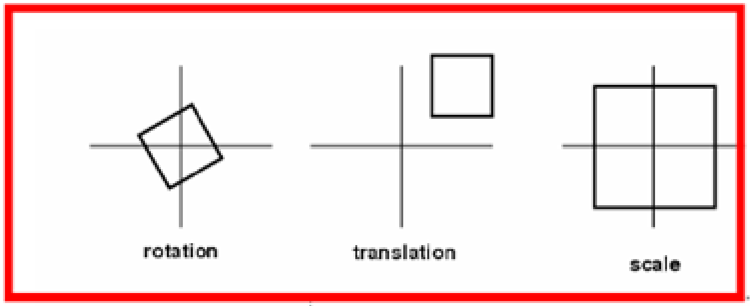
$$\begin{bmatrix} p' \\ 1 \end{bmatrix} \sim \begin{bmatrix} A & b \\ c^T & 1 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix}$$

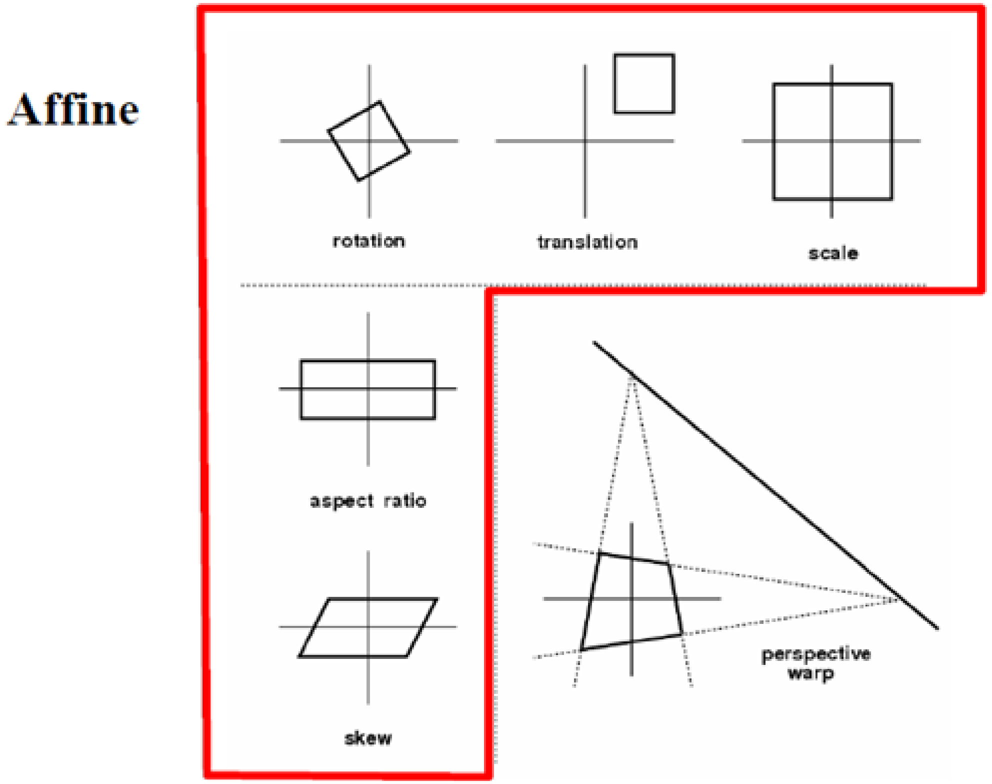
matrix form

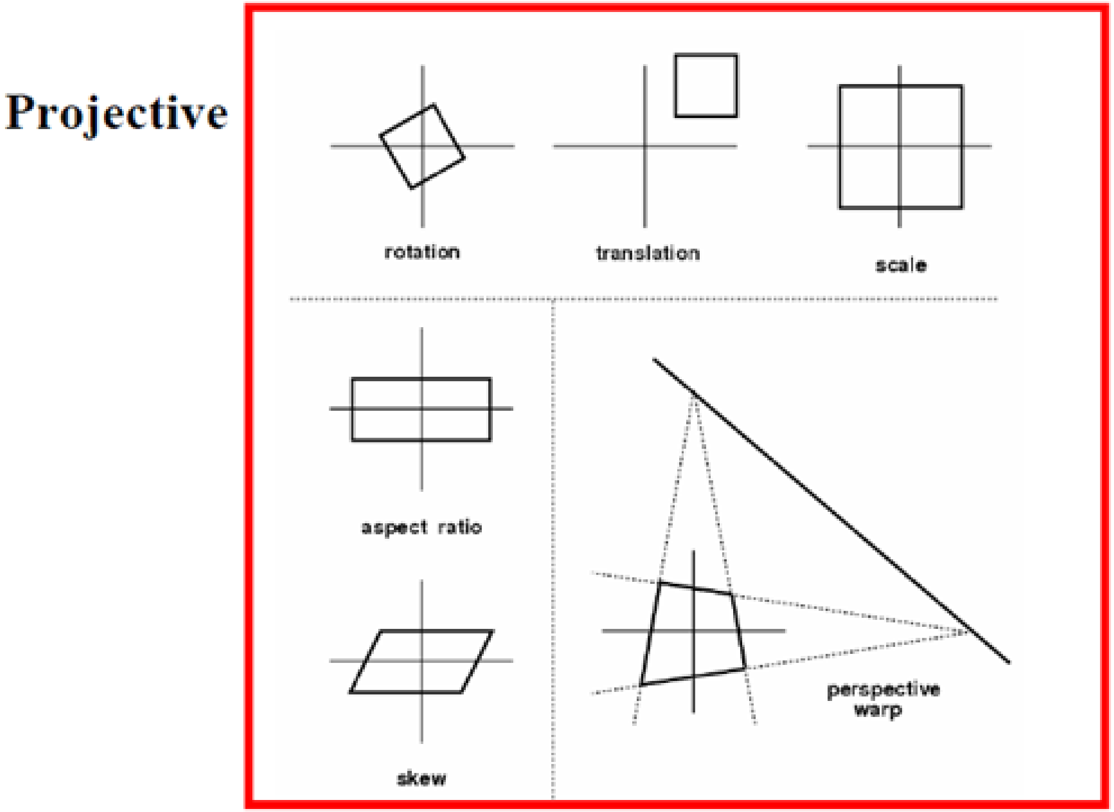
Euclidean



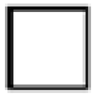
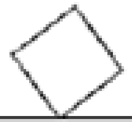
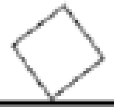
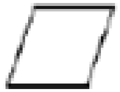
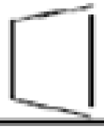
Similarity



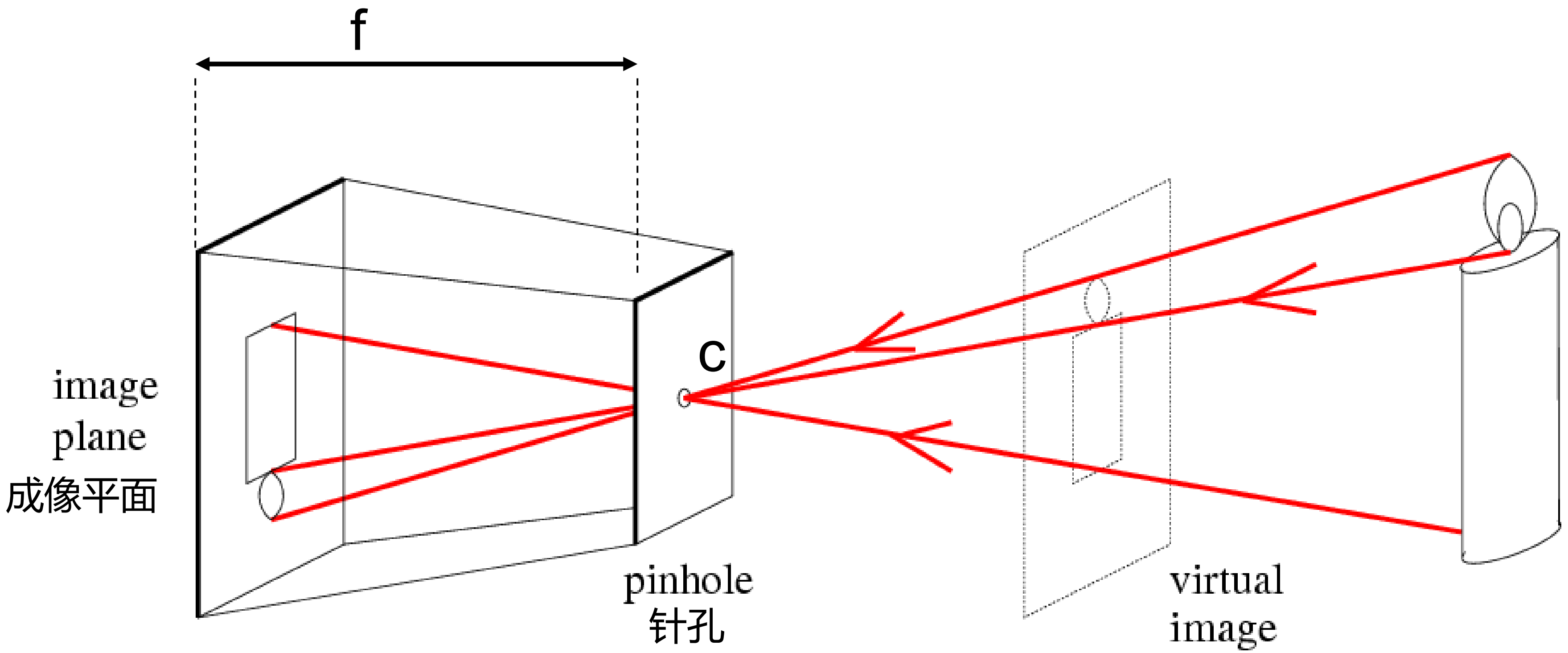

Affine
rotation
translation
scale
aspect ratio
skew
perspective warp


Projective
rotation
translation
scale
aspect ratio
skew
perspective warp

成像几何基础

Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\begin{bmatrix} I \mid t \end{bmatrix}_{2\times 3}$	2	orientation $+\cdots$	
rigid (Euclidean)	$\begin{bmatrix} R \mid t \end{bmatrix}_{2\times 3}$	3	lengths $+\cdots$	
similarity	$\begin{bmatrix} sR \mid t \end{bmatrix}_{2\times 3}$	4	angles $+\cdots$	
affine	$\begin{bmatrix} A \end{bmatrix}_{2\times 3}$	6	parallelism $+\cdots$	
projective	$\begin{bmatrix} H \end{bmatrix}_{3\times 3}$	8	straight lines	

针孔相机



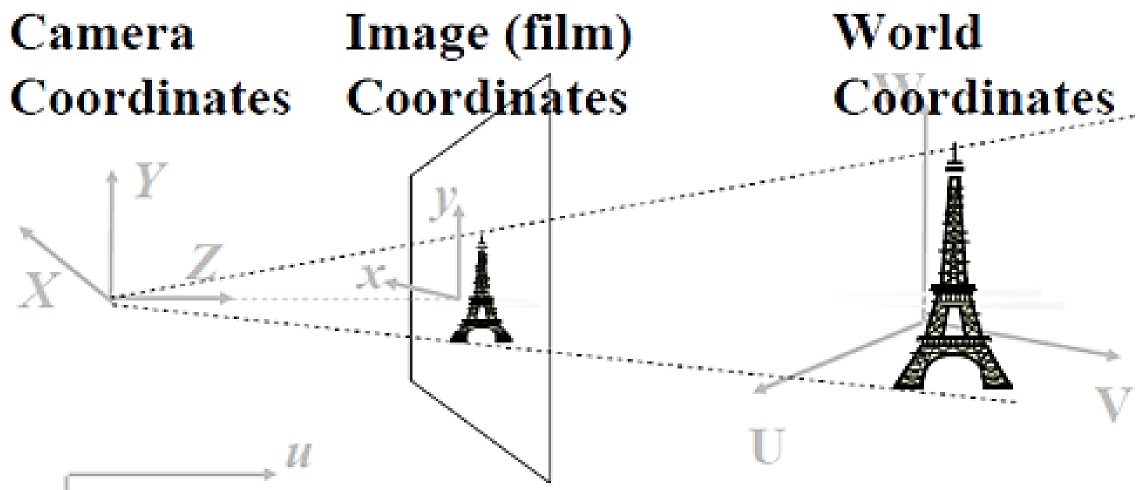
f = 焦距

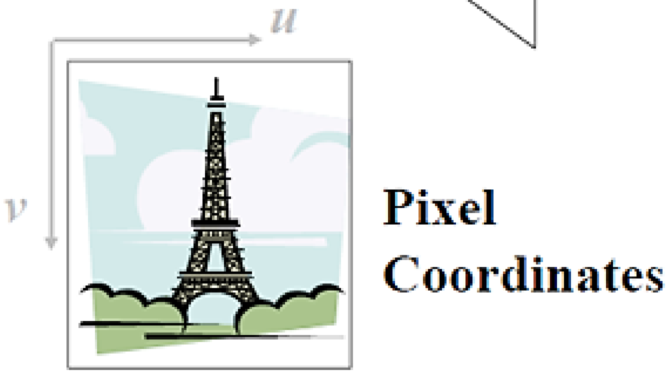
c = 相机中心

相机坐标 (X,Y,Z)

像平面坐标 (x,y)

世界坐标或物体坐标 (U,V,W)





数字图像用(u,v)来表示，不能混淆像平面和数字图像这两个概念，同一个像通过平移、拉伸等，可以得到不同的数字图像(u,v)。

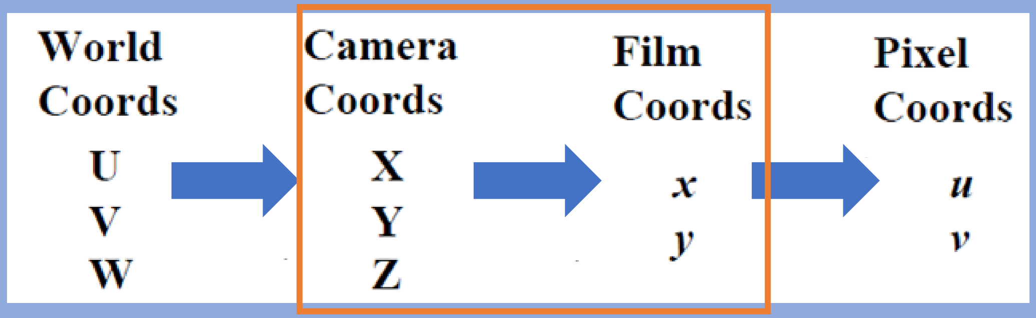
- 试想像一下，很多游客同时在不同角度拍摄Eiffel Tower(埃菲尔铁塔)，该如何用数学的方法来描述这一过程呢？

- 首先要解决的问题就是定位，或者说坐标选定的问题。
  埃菲尔铁塔只有一座，如果按经、纬度来刻画，它的坐标是唯一确定的，但游客显然不关心这一点，他(她)只按自己的喜好选择角度和位置。 因此，**物体(景物)有物体的坐标系统，相机有相机的坐标系统**，即便同一个相机，当调整参数时，在同样的位置、相同的角度，也可能得到不同的图像。

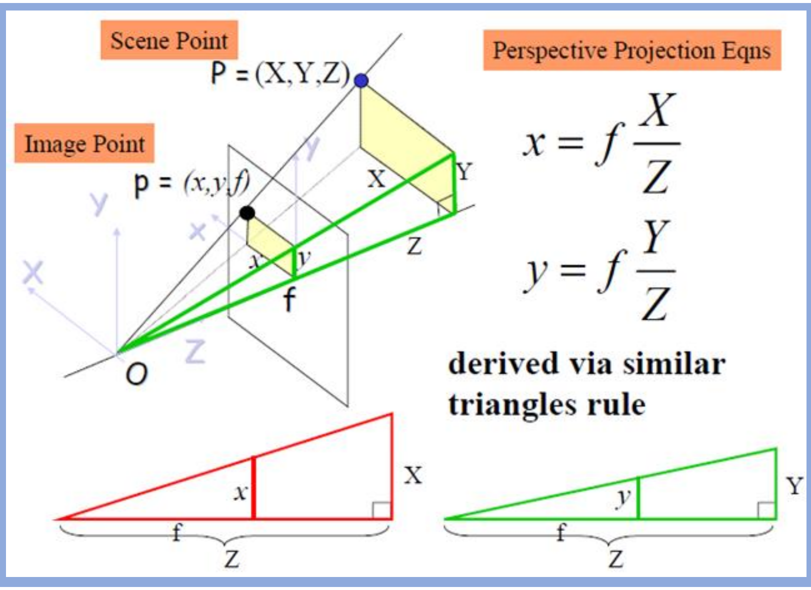
- 为了统一描述，有必要引入世界坐标(或物体坐标)、相机坐标和像平面坐标。

【问题】如何将3D世界坐标系中的点变换到相机坐标系中，然后经透视投影，变成2D像平面上的点(x,y)?

计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2293932.html



红框标注的部分是3D物体到2D像平面的透视投影



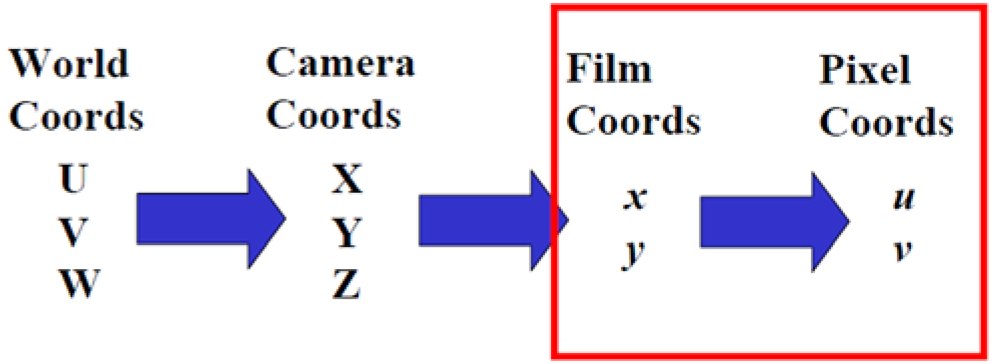
显然，OP上的任一点的像都是p(x,y)，为了描述这一关系，需要引入齐次坐标。

By convention, we specify that given (x’,y’,z’) , we can recover the 2D point (x, y) as x=x’/z’; y=y’/z’. Note that
(x,y) = (x,y,1) = (2x, 2y, 2) = (k x, ky, k)

透视投影的过程可以描述为:

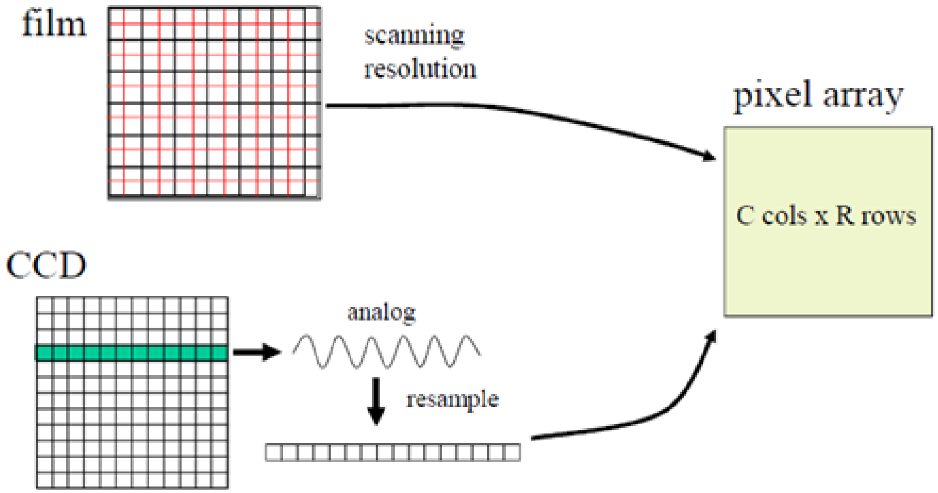
$$x = f\frac{X}{Z} \quad y = f\frac{Y}{Z} \Longleftrightarrow \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

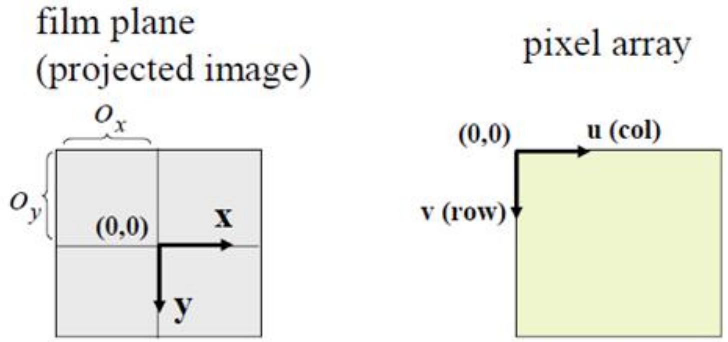
计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2293932.html



Affine Transformation

主要讨论如何从像平面(x,y)变换到数字图像(u,v)，即从像平面(Film Coords)到像素(Pixel Coords)。





$$u = f\frac{X}{Z} + o_x \qquad v = f\frac{Y}{Z} + o_y$$

o_x and o_y called image center or principle point

$$u = \frac{1}{s_x} f \frac{X}{Z} + o_x \qquad v = \frac{1}{s_y} f \frac{Y}{Z} + o_y$$

Adding the intrinsic parameters into the perspective projection matrix:

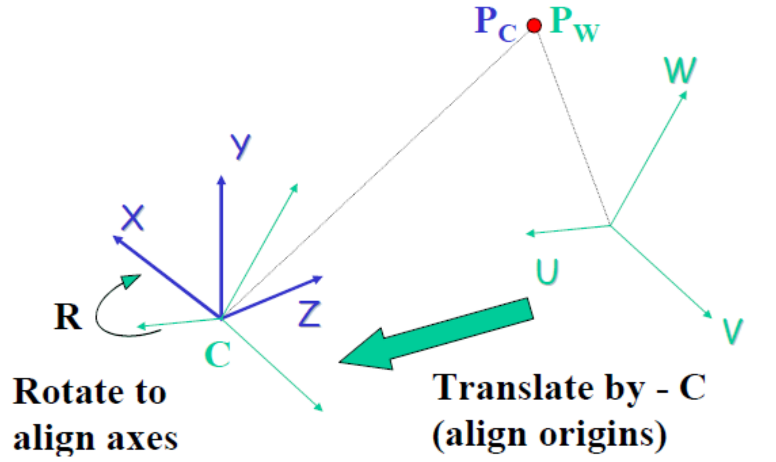
$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} f/s_x & 0 & o_x & 0 \\ 0 & f/s_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2294011.html

World Coords $\begin{pmatrix}U\\V\\W\end{pmatrix}$ → Camera Coords $\begin{pmatrix}X\\Y\\Z\end{pmatrix}$ → Film Coords $\begin{pmatrix}x\\y\end{pmatrix}$ → Pixel Coords $\begin{pmatrix}u\\v\end{pmatrix}$

红框部分描述的是从物体的坐标(称为世界坐标)到相机坐标变换的过程，它是一种刚体运动，可以用平移、旋转来描述。

R：世界坐标系坐标轴相对于相机坐标系坐标轴



上图表示的是从世界坐标变换到相机坐标：$\mathbf{P}_C = \mathbf{R}(\mathbf{P}_W - \mathbf{C})$

写成矩阵形式

$$\begin{pmatrix}X\\Y\\Z\\1\end{pmatrix} = \begin{pmatrix}r_{11} & r_{12} & r_{13} & 0\\ r_{21} & r_{22} & r_{23} & 0\\ r_{31} & r_{32} & r_{33} & 0\\ 0 & 0 & 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0 & 0 & -c_x\\ 0 & 1 & 0 & -c_y\\ 0 & 0 & 1 & -c_z\\ 0 & 0 & 0 & 1\end{pmatrix}\begin{pmatrix}U\\V\\W\\1\end{pmatrix}$$

$$\begin{pmatrix}X\\Y\\Z\\1\end{pmatrix} = \begin{pmatrix}r_{11} & r_{12} & r_{13} & 0\\ r_{21} & r_{22} & r_{23} & 0\\ r_{31} & r_{32} & r_{33} & 0\\ 0 & 0 & 0 & 1\end{pmatrix}\begin{pmatrix}U\\V\\W\\1\end{pmatrix} \qquad \mathbf{P_C} = \mathbf{R}\,\mathbf{P_W}$$

简化：假设世界坐标系和相机坐标系的原点重合，则变换只剩下旋转

计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2293932.html0

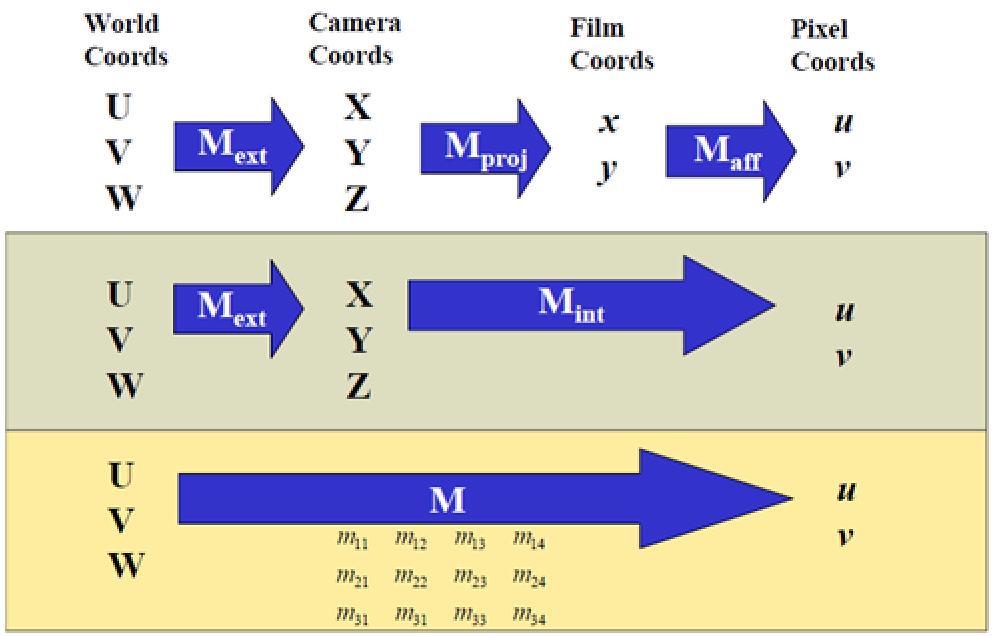
$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} f/s_x & 0 & o_x & 0 \\ 0 & f/s_y & o_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\begin{pmatrix} u' \\ v' \\ w' \end{pmatrix} = \underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{pmatrix}}_{\mathbf{M}_{\mathbf{aff}}} \underbrace{\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{\mathbf{M}_{\mathbf{proj}}} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\mathbf{u} = \mathbf{M}_{\mathbf{int}}\, \mathbf{P}_{\mathbf{C}} = \mathbf{M}_{\mathbf{aff}}\, \mathbf{M}_{\mathbf{proj}}\, \mathbf{P}_{\mathbf{C}}$$

计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2294011.html

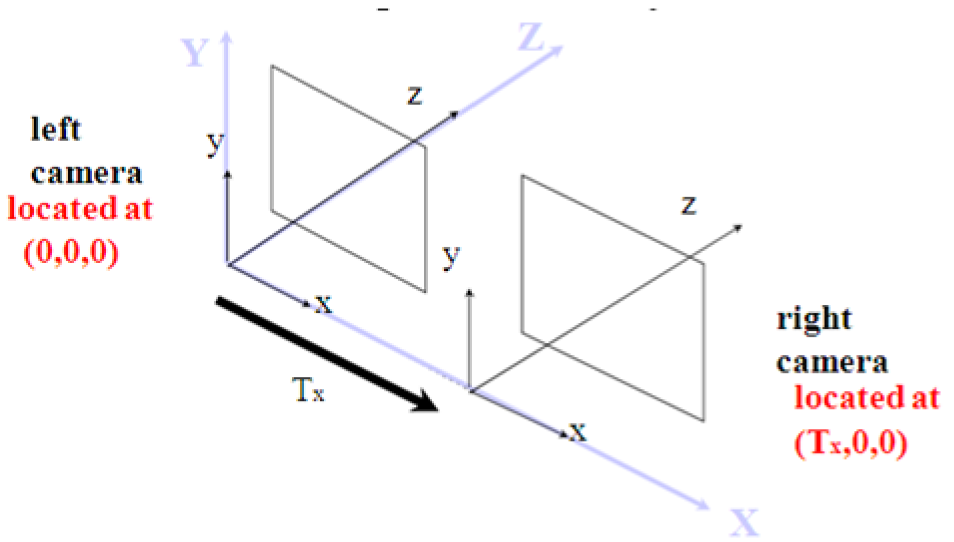
所以，整个相机的成像矩阵可以分解为：

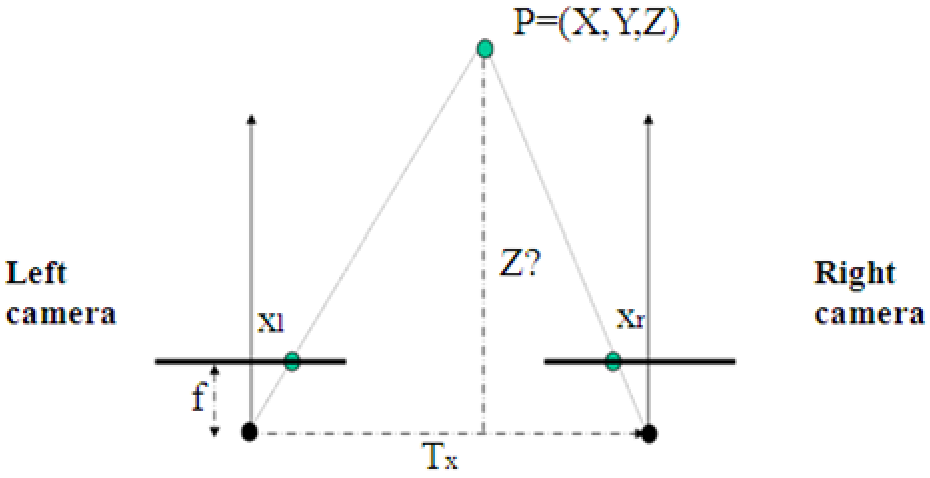


$$\begin{bmatrix}u\\v\\1\end{bmatrix} \sim \underbrace{\underbrace{\underbrace{\overset{\text{Film plane to pixels}}{\begin{bmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\0&0&1\end{bmatrix}}}_{\mathbf{M}_{\mathbf{aff}}}\underbrace{\overset{\text{Perspective projection}}{\begin{bmatrix}f&0&0&0\\0&f&0&0\\0&0&1&0\end{bmatrix}}}_{\mathbf{M}_{\mathbf{proj}}}}_{\mathbf{M}_{\mathbf{int}}}\underbrace{\overset{\text{World to camera}}{\begin{bmatrix}r_{11}&r_{12}&r_{13}&t_x\\r_{21}&r_{22}&r_{23}&t_y\\r_{31}&r_{32}&r_{33}&t_z\\0&0&0&1\end{bmatrix}}}_{\mathbf{M}_{\mathbf{ext}}}}_{\mathbf{M}}\begin{bmatrix}U\\V\\W\\1\end{bmatrix}$$

计算机视觉基础2——相机成像的几何描述. https://www.cnblogs.com/gemstone/archive/2011/12/19/2294011.html

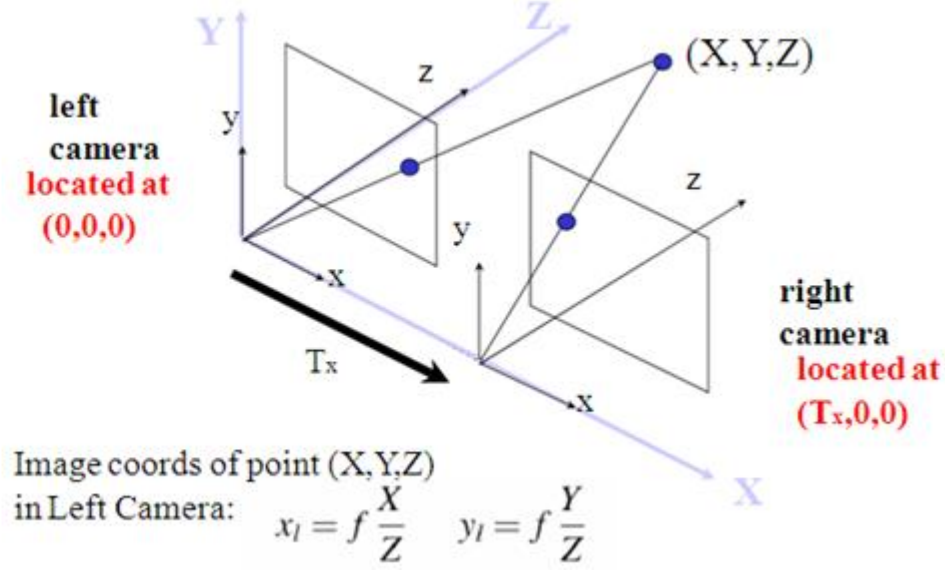
假设两个相机的内部参数一致，如焦距、镜头等等

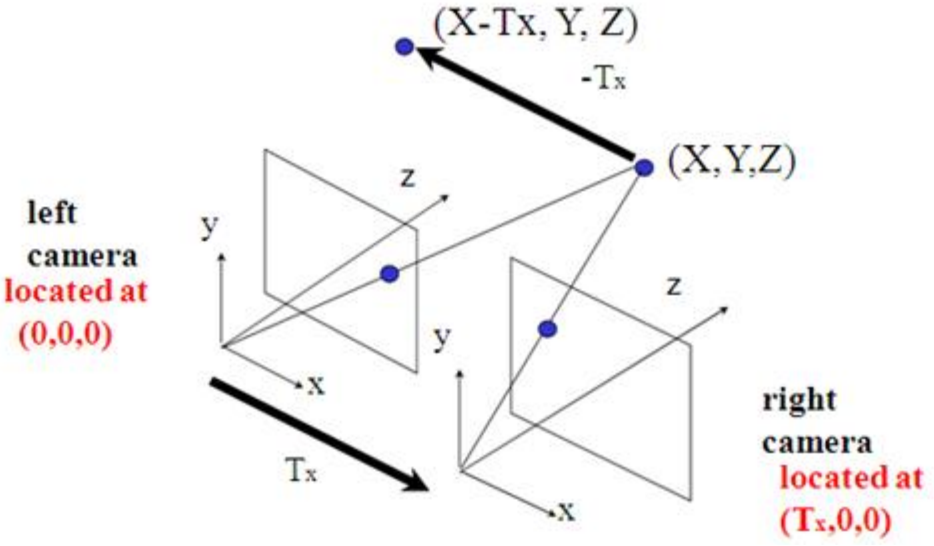


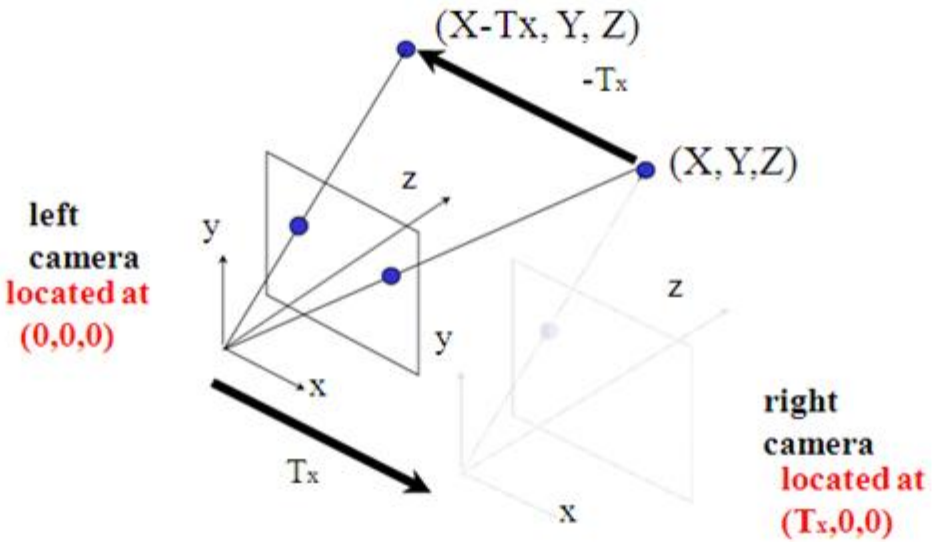


我们将会在后面专门开设一节课来讲3D视觉，到时候会介绍这部分内容

双目视觉







Tx一般称为基线(baseline)，根据三角形相似关系，很容易得出空间中的一点P(X,Y,Z)分别在左右像平面上的投影坐标。

因此，左相机像平面像点的坐标为

xl=fX/Z　　　　yl=fY/Z

右相机平面像点的坐标为

xr=f(X−Tx)/Z　　yr=fY/Z

因此，左相机像平面像点的坐标为

xl=fX/Z　　　　yl=fY/Z

右相机平面像点的坐标为

xr=f(X−Tx)/Z　　yr=fY/Z

Stereo Disparity

$$d = x_l - x_r = f\frac{X}{Z} - \left(f\frac{X}{Z} - f\frac{T_x}{Z}\right)$$

$$d = \frac{f\,T_x}{Z}$$

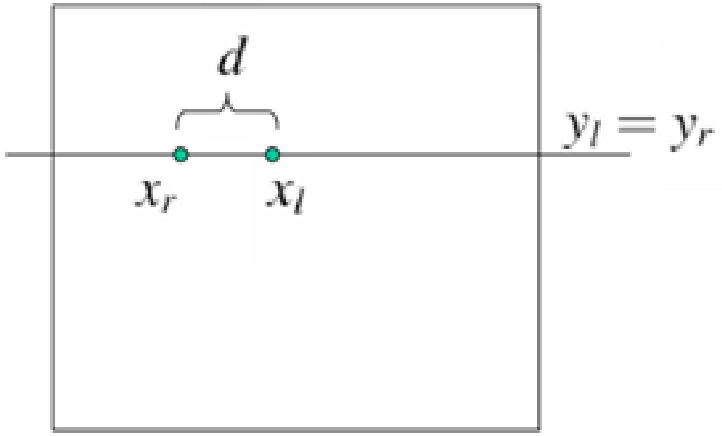
depth　baseline

$$Z = \frac{f\,T_x}{d}$$

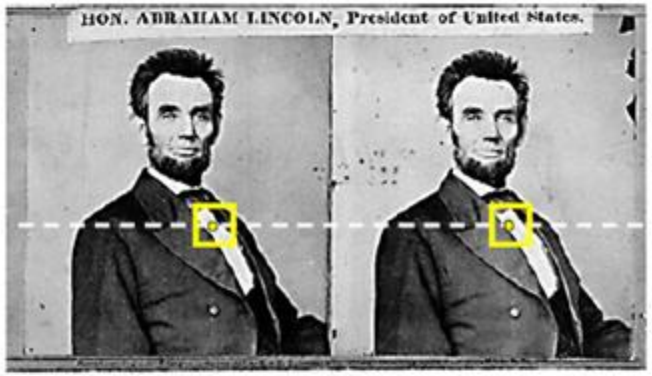
disparity

Important equation!

深度信息Z和视差(Disparity / Parallax) d成反比，



如何估计d?

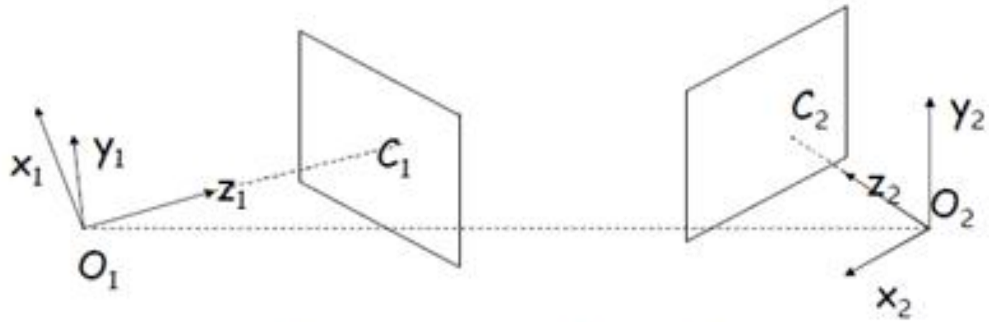


Corresponding features are constrained to lie along conjugate epipolar lines (on the same row in the case of our simple setup).

双目视觉-非平行光轴情况

以上双目为平行光轴情况，但如果非平行光轴呢?

一般的立体成像关系：两个相机的坐标无任何约束关系，相机的内部参数可能不同，甚至是未知的。要刻画这种情况下的两幅图像之间的对应关系，需要引入**对极几何**以及几个重要的概念——对极矩阵(Epipolar Matrix)和基本矩阵(Fundamental Matrix)等。



In general, the cameras may be related by an arbitrary transformation (R,T)

Epipolar Matrix

In general, intrinsic camera parameters may be different, and even unknown

Fundamental Matrix

双目视觉-非平行光轴情况

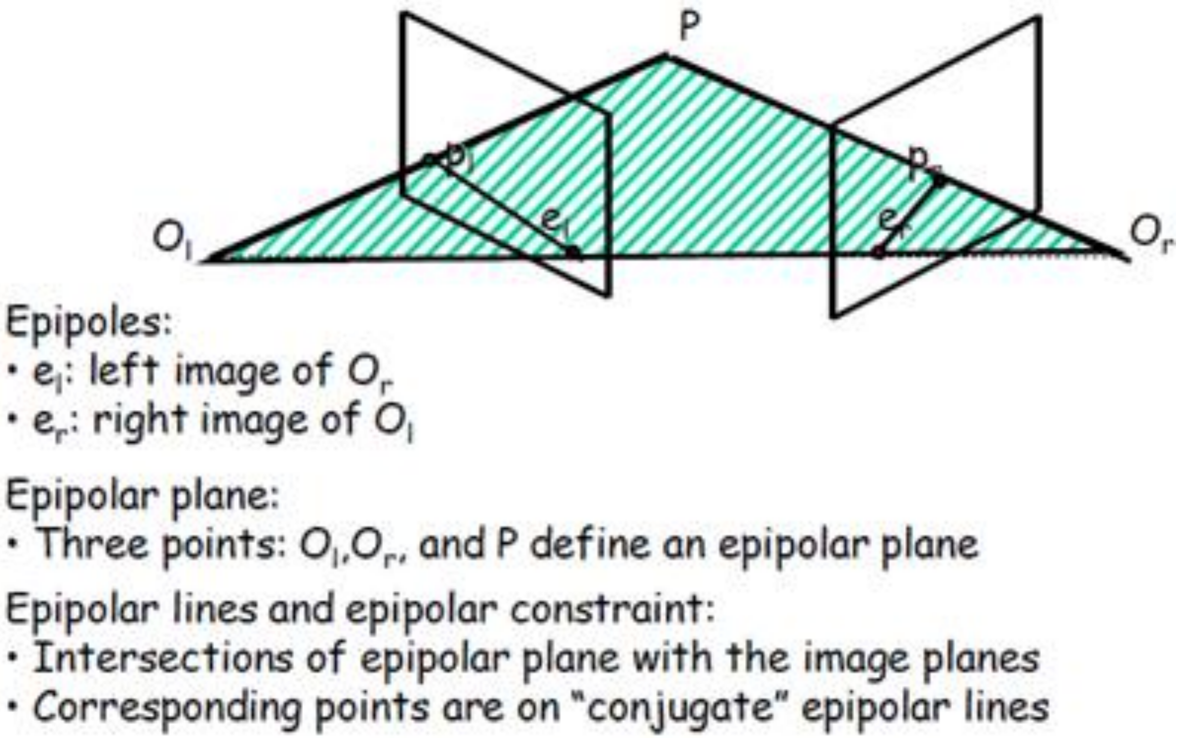
对极几何中的重要概念（参考下图）：

极点：极点e_l:右相机坐标原点在左像平面上的像；极点e_r:左相机坐标原点在右像平面上的像

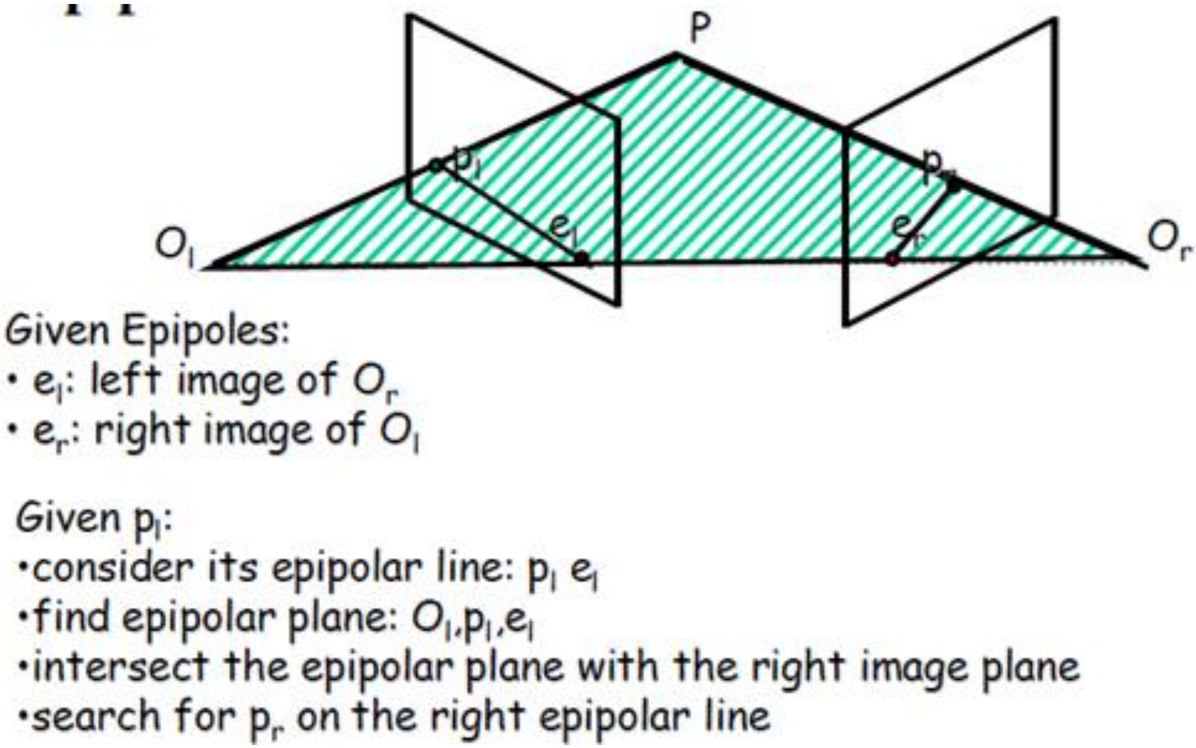
极平面：由两个相机坐标原点O_l、O_r和物点P组成的平面

级线：极平面与两个像平面的交线，即p_le_l和p_re_r

级线约束：两极线上点的对应关系



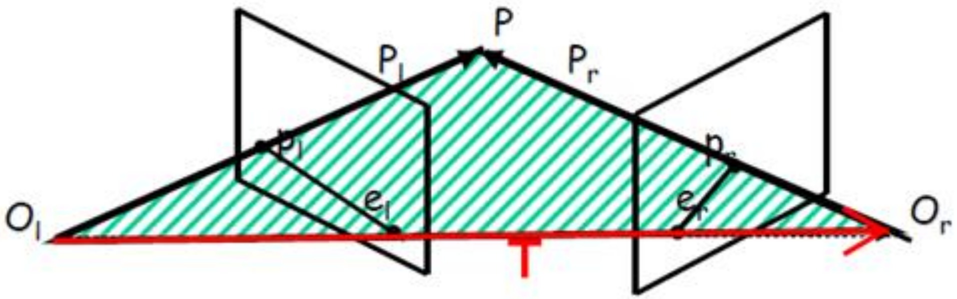
几个概念



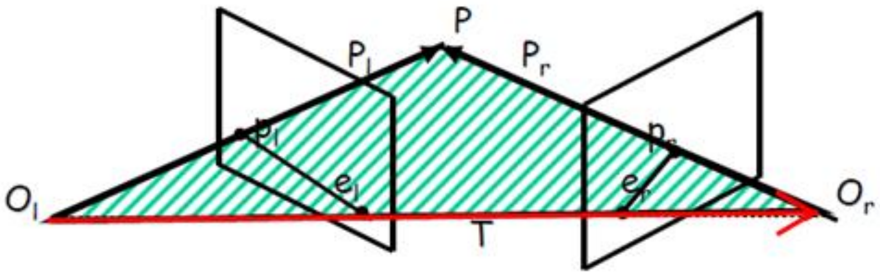
对极约束

双目视觉-非平行光轴情况

两个相机坐标系之间的关系为



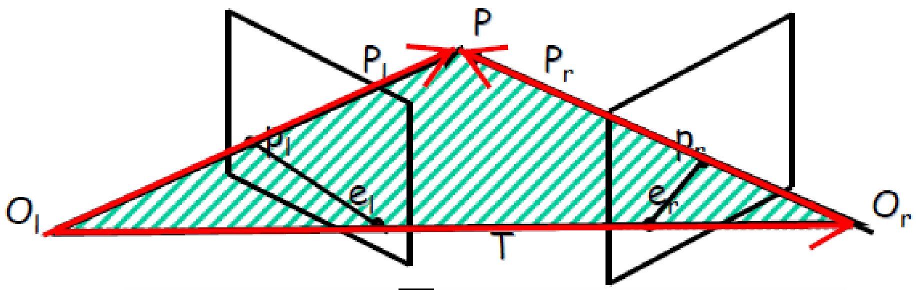
$$P_r = R(P_l - T)$$



$$P_r = R(P_l - T)$$

$$P_l - T = R^{-1} P_r = R^T P_r$$

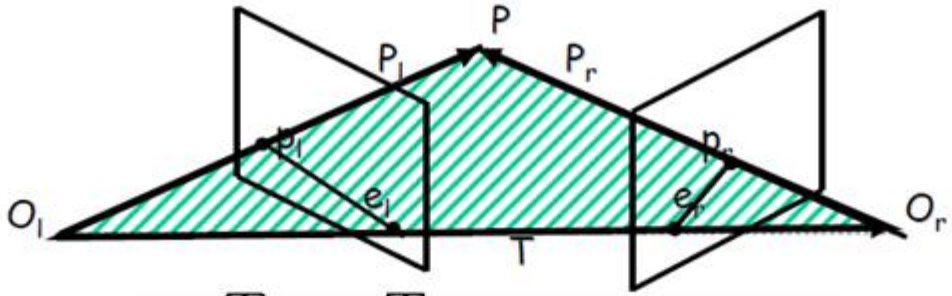
Epipolar constraint: P_l, T and P_l - T are coplanar:



$$(P_l - T)^T \cdot T \times P_l = 0$$

$$P_l - T = R^T P_r \Rightarrow (R^T P_r)^T \cdot T \times P_l = 0$$

Epipolar constraint: P_l, T and P_l - T are coplanar:

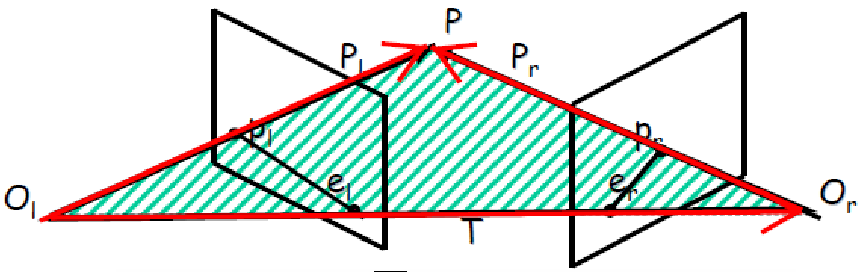


$$(R^T P_r)^T \cdot T \times P_l = 0$$

$$(P_r^T R) \cdot (T \times P_l) = 0$$

双目视觉-非平行光轴情况

Epipolar constraint: P_l, T and P_l - T are coplanar:



$$(P_l - T)^T \cdot T \times P_l = 0$$

$$P_l - T = R^T P_r \quad \Rightarrow (R^T P_r)^T \cdot T \times P_l = 0$$

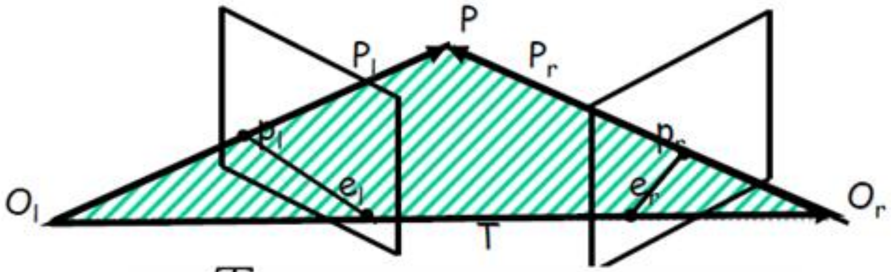
$$T \times P_l = \begin{vmatrix} i & j & k \\ T_x & T_y & T_z \\ P_{l_x} & P_{l_y} & P_{l_z} \end{vmatrix}$$

$$T \times P_l = (T_y P_{l_z} - T_z P_{l_y}) i + (T_z P_{l_x} - T_x P_{l_z}) j + (T_x P_{l_y} - T_y P_{l_x}) k$$

$$T \times P_l = S P_l = \begin{bmatrix} 0 & -T_z & T_y \\ T_z & 0 & -T_x \\ -T_y & T_x & 0 \end{bmatrix} \begin{bmatrix} P_{l_x} \\ P_{l_y} \\ P_{l_z} \end{bmatrix} = \begin{bmatrix} T_y P_{l_z} - T_z P_{l_y} \\ T_z P_{l_x} - T_x P_{l_z} \\ T_x P_{l_y} - T_y P_{l_x} \end{bmatrix}$$

S has rank 2 ; it depends only on T

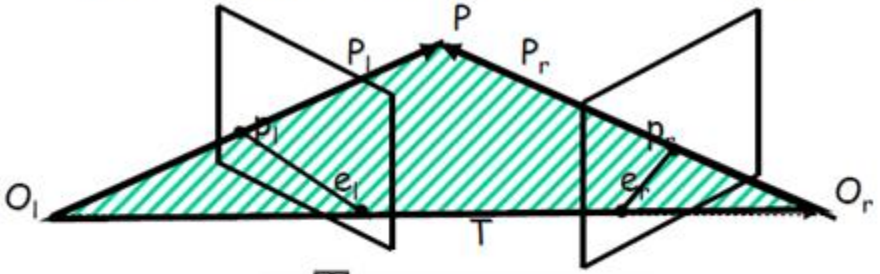
Epipolar constraint: P_l, T and P_l - T are coplanar:



$$(P_r^T R) \cdot (T \times P_l) = 0$$

$$P_r^T R S P_l = 0$$

Epipolar constraint: P_l, T and P_l - T are coplanar:



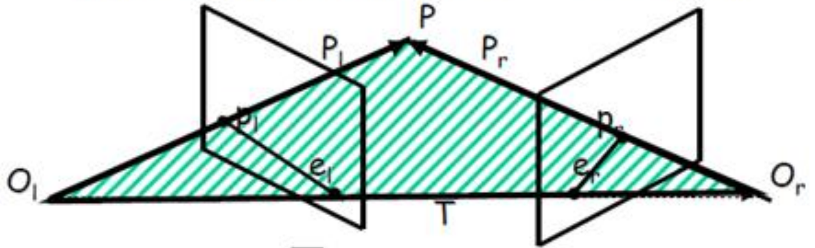
$$P_r^T R S P_l = 0$$

Essential Matrix:

$$E = RS \qquad P_r^T E P_l = 0$$

双目视觉-非平行光轴情况

Epipolar constraint: P_l, T and P_l - T are coplanar:



$$P_r^T RSP_l = 0$$

Essential Matrix:

$$E = RS \qquad P_r^T E P_l = 0$$

显然，左像点p_l和右像点p_r是通过矩阵E=RS来约束的，我们称矩阵E为本质矩阵(Essential Matrix)，它的基本性质有：

- has rank 2(秩为2)
- depends only on the EXTRINSIC Parameters (R & T)(仅依赖于外部参数R和T)

结合成像的几何关系

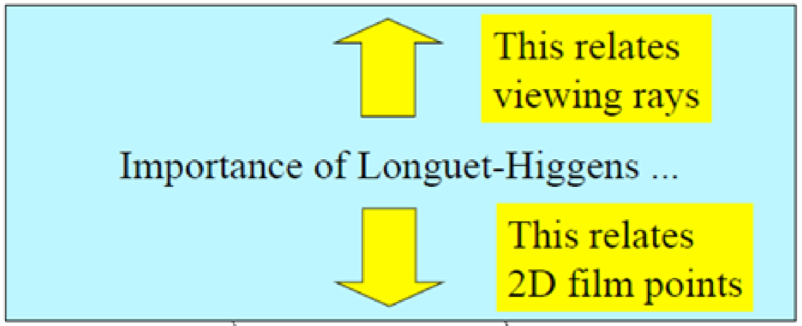
$$P_r^T E P_l = 0$$

$$p_l = \frac{f_l}{Z_l} P_l \qquad p_r = \frac{f_r}{Z_r} P_r$$

$$\left(\cancel{\frac{Z_r}{f_r}} p_r\right)^T E \left(\cancel{\frac{Z_l}{f_l}} p_l\right) = 0$$

$$\boxed{p_r^T E p_l = 0}$$

Longuet-Higgins equation $P_r^T E P_l = 0$



$$\boxed{p_r^T E p_l = 0}$$

双目视觉-非平行光轴情况

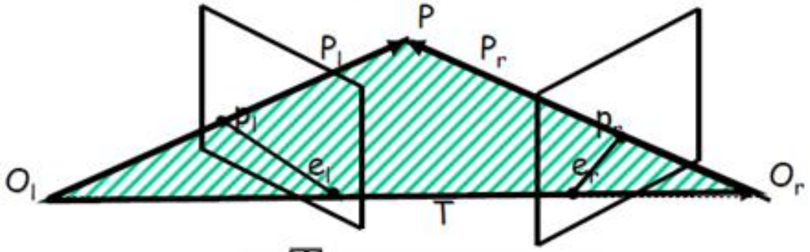
注意大小写的区别，大小表示物点矢量，小与表示像点矢量。
像平面上的一点可以看作：

- (u,v) a 2D film point (局限于像平面上来考虑)
- (u,v,f) a 3D point on film plane (相机坐标系中来考虑)
- k(u,v,f) a viewing ray into the scene (透过像点和原点射线上点的像，相机坐标系中来考虑)
- k(X, Y, Z) a ray through point P in the scene (在世界坐标系中来考虑)

设l为像平面上的一直线：au+bv+c=0
由点线结合关系可得：

$$\tilde{p} = \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \quad \tilde{l} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \boxed{\tilde{p}^T \tilde{l} = \tilde{l}^T \tilde{p} = 0}$$

Epipolar constraint: P_l, T and P_l - T are coplanar:



$$P_r^T R S P_l = 0$$

Essential Matrix:

$$E = RS \qquad P_r^T E P_l = 0$$

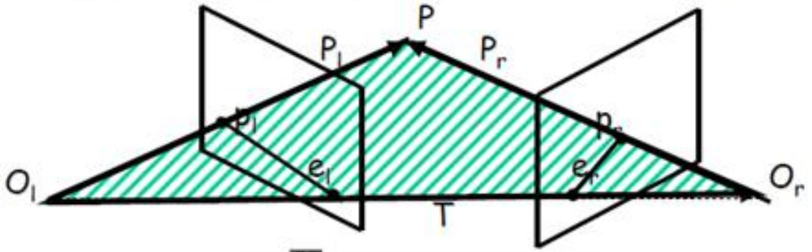
$$p_r^T E p_l = 0$$

$$\tilde{l}_r = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

p_r belongs to epipolar line in the right image defined by

$$\tilde{l}_r = E p_l$$

Epipolar constraint: P_l, T and P_l - T are coplanar:



$$P_r^T R S P_l = 0$$

Essential Matrix:

$$E = RS \qquad P_r^T E P_l = 0$$

$$p_r^T E p_l = 0$$

$$\tilde{l}_r = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

p_r belongs to epipolar line in the right image defined by

$$\tilde{l}_r = E p_l$$

这样就可以用几何的观点来解释上述方程：左像平面上的一点pl乘以本质矩阵E，结果为一条直线，该直线就是pl的极线，且过pl在右像平面上的对应点pr

同理，

$$p_r^T E p_l = 0$$

$$\tilde{l}_l^T = \begin{bmatrix} a \\ b \\ c \end{bmatrix}^T$$

p_l belongs to epipolar line in the left image defined by

$$\tilde{l}_l = E^T p_r$$

- Remember: epipoles belong to the epipolar lines

$$e_r^T E p_l = 0 \qquad p_r^T E e_l = 0$$

- And they belong to all the epipolar lines, i.e., any p_l or p_r, s.t.,

$$e_r^T E = 0 \qquad E e_l = 0$$

关于本质矩阵的关系总结如下：

Longuet-Higgins equation $p_r^T E p_l = 0$

Epipolar lines: $\tilde{p_r}^T \tilde{l_r} = 0 \qquad \tilde{p_l}^T \tilde{l_l} = 0$

$\tilde{l_r} = E p_l \qquad \tilde{l_l} = E^T p_r$

Epipoles: $e_r^T E = 0 \qquad E e_l = 0$

双目视觉-非平行光轴情况

本质矩阵采用的是相机的外部参数，也就是说采用相机坐标(The essential matrix uses CAMERA coordinates)，如果要分析数字图像，则要考虑坐标(u,v)，此时需要用到内部参数

Pixel coord (row,col)　　Affine transform matrix　　Camera (film) coord

$$\bar{p}_l = M_l p_l \qquad p_l = M_l^{-1}\bar{p}_l$$

$$\bar{p}_r = M_r p_r \qquad p_r = M_r^{-1}\bar{p}_r$$

$$p_l = M_l^{-1}\bar{p}_l$$
$$p_r = M_r^{-1}\bar{p}_r \qquad p_r^T E p_l = 0$$

$$(M_r^{-1}\bar{p}_r)^T E (M_l^{-1}\bar{p}_l) = 0$$

$$\bar{p}_r^T (M_r^{-T} E M_l^{-1}) \bar{p}_l = 0$$

$$\boxed{\bar{p}_r^T F \bar{p}_l = 0}$$

矩阵F称为基本矩阵:$F=M_r^{-T}RSM_l^{-1}$

- has rank 2
- depends on the INTRINSIC and EXTRINSIC Parameters (f, etc ; R & T)

Analogous to essential matrix. The fundamental matrix also tells how pixels (points) in each image are related to epipolar lines in the other image.

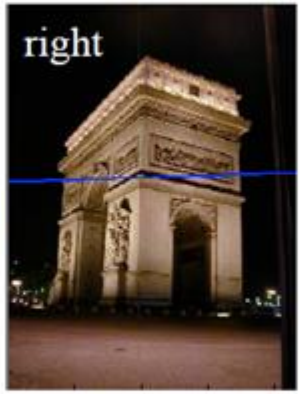
$$F = \begin{pmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{pmatrix} \begin{pmatrix} 343.53 \\ 221.70 \\ 1.0 \end{pmatrix}$$

x = 343.5300 y = 221.7005

$$\begin{matrix} 0.0001 \\ 0.0045 \\ -1.1942 \end{matrix} \rightarrow \begin{matrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{matrix}$$

normalize so sum of squares
of first two terms is 1 (optional)

$$\begin{matrix} 0.0295 \\ 0.9996 \\ -265.1531 \end{matrix}$$



$$(205.5526 \quad 80.5 \quad 1.0) \begin{pmatrix} -0.00310695 & -0.0025646 & 2.96584 \\ -0.028094 & -0.00771621 & 56.3813 \\ 13.1905 & -29.2007 & -9999.79 \end{pmatrix}$$

$L = (0.0010 \quad -0.0030 \quad -0.4851)$

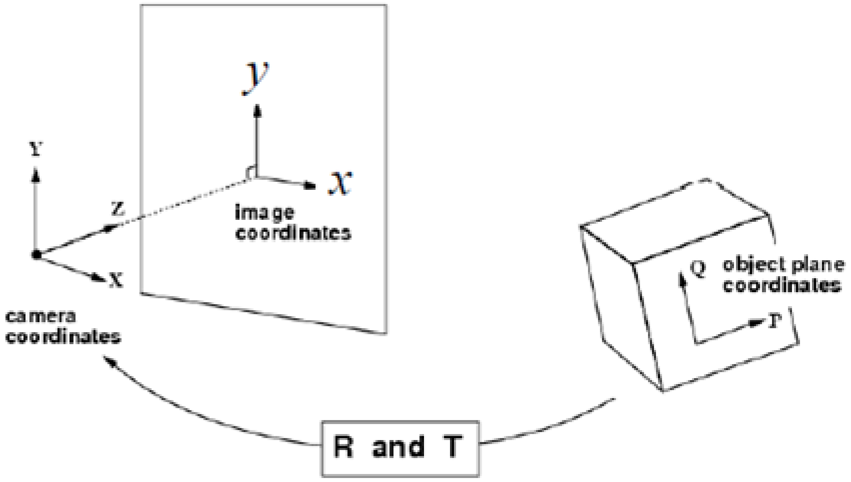
$\rightarrow (0.3211 \quad -0.9470 \; -151.39)$

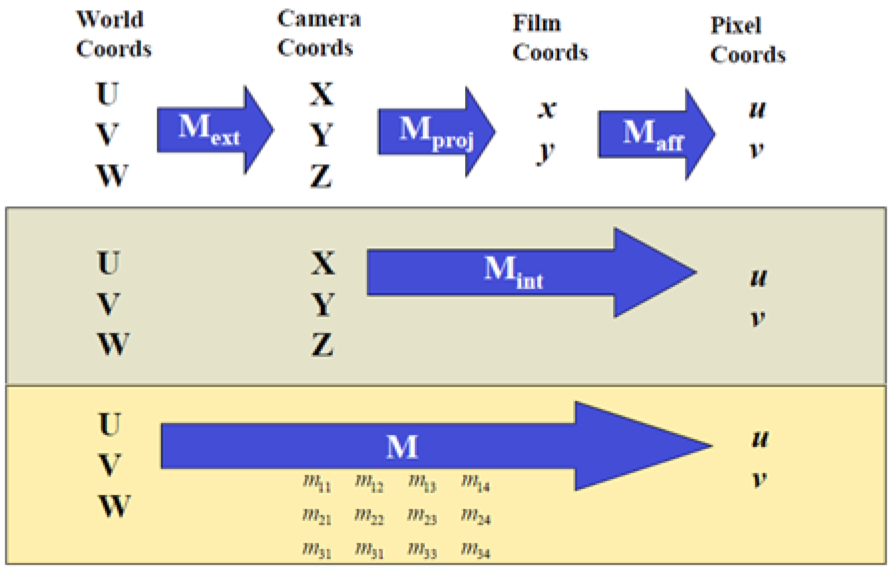
x = 205.5526 y = 80.5000

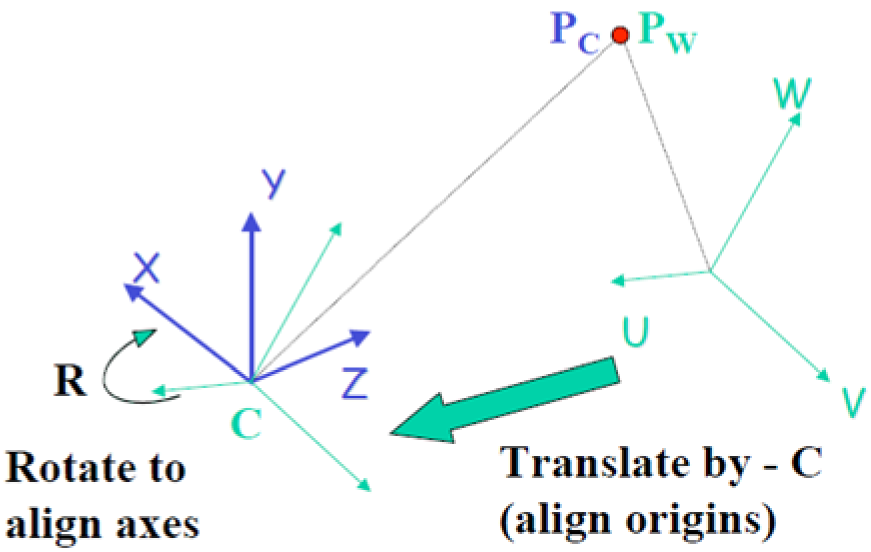
$L = (0.3211 \quad -0.9470 \quad -151.39)$

例子

三维物体的成像:







$$
\begin{aligned}
P_C &= R\,(P_W - C) \\
&= R\,P_W + T
\end{aligned}
$$

Line preserving

Definition:

A 2D *homography* is an invertible mapping h from P^2 to itself such that three points x_1,x_2,x_3 lie on the same line if and only if $h(x_1)$,$h(x_2)$,$h(x_3)$ do.

Theorem:

A mapping $h: P^2 \rightarrow P^2$ is a homography if and only if there exist a non-singular 3x3 matrix **H** such that for any point in P^2 represented by a vector x it is true that $h(x)=\mathbf{H}x$

Definition: Homography

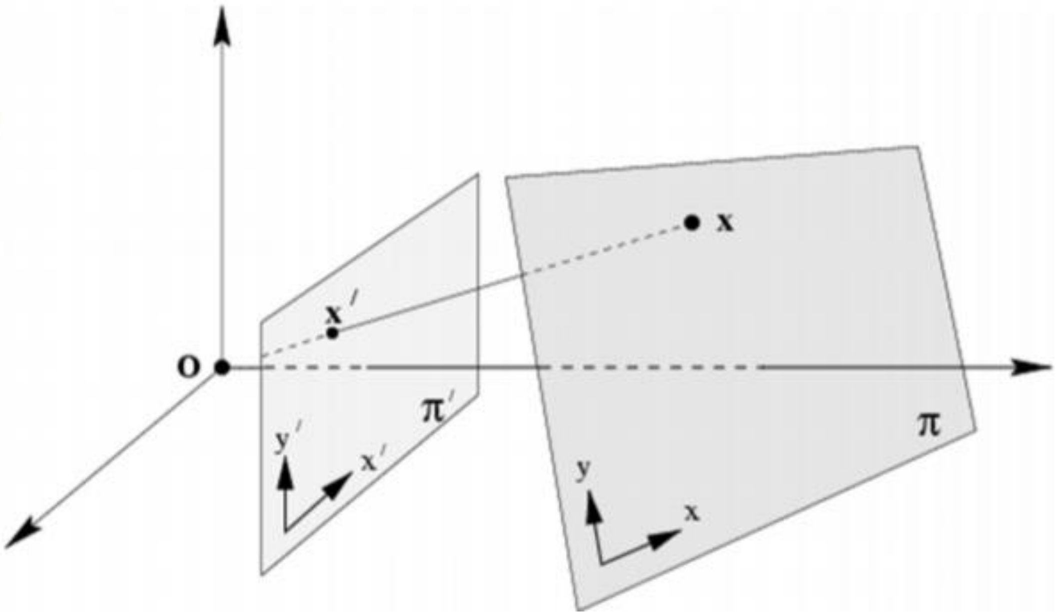
$$\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \text{or} \quad \mathrm{x}' = \mathbf{H}\,\mathrm{x}$$

8DOF

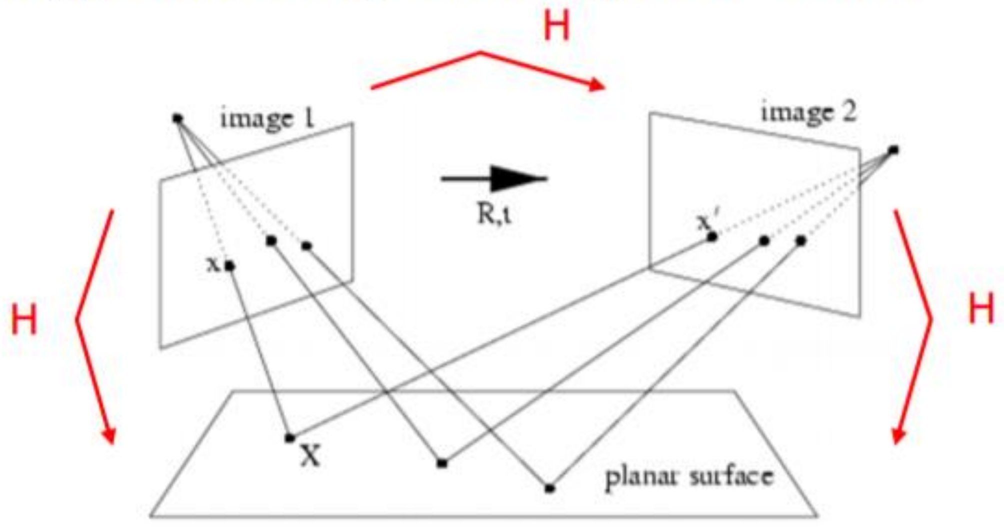
Homography=projective transformation=projectivity=collineation

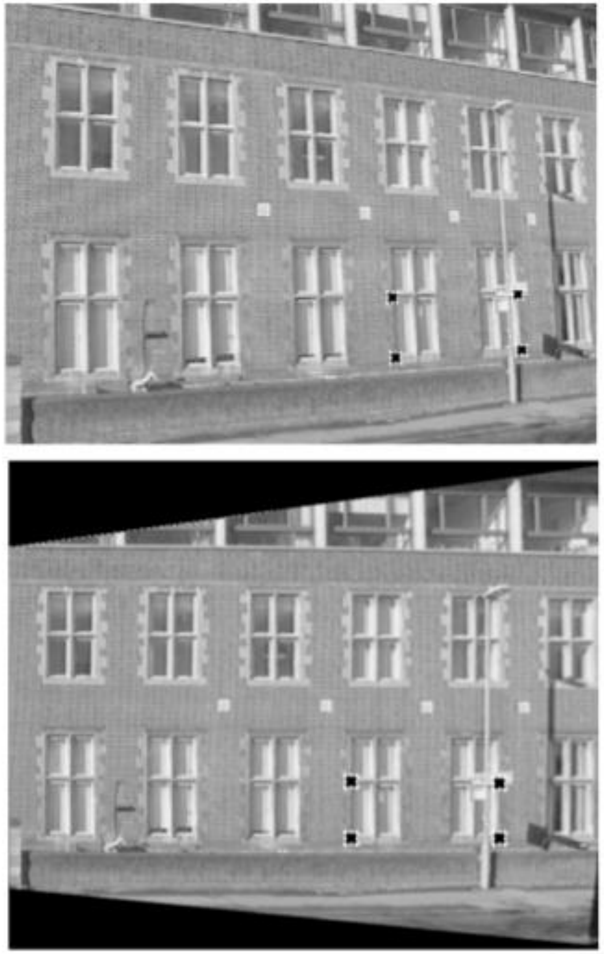
- Note: homographies are not restricted to P^2
- General definition:
  A homography is a non-singular, line preserving, projective mapping h: $P^n \rightarrow P^n$.
  It is represented by a square (n + 1)-dim matrix with $(n + 1)^2$-1 DOF

- Now back to the 2D case…
- Mapping between planes



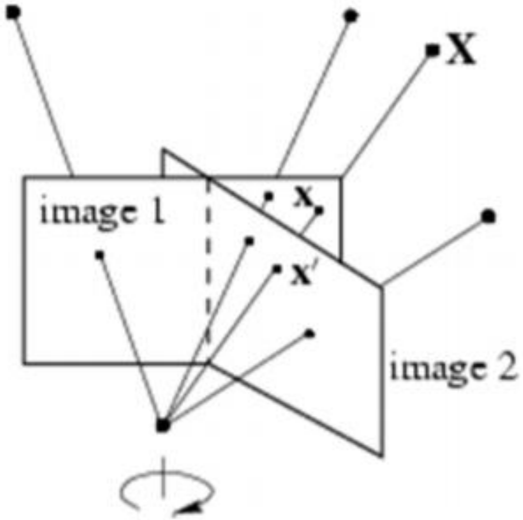
Rotating/translating camera, planar world

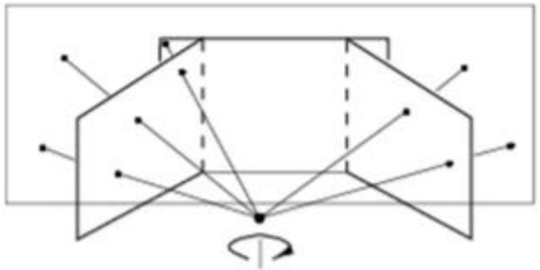


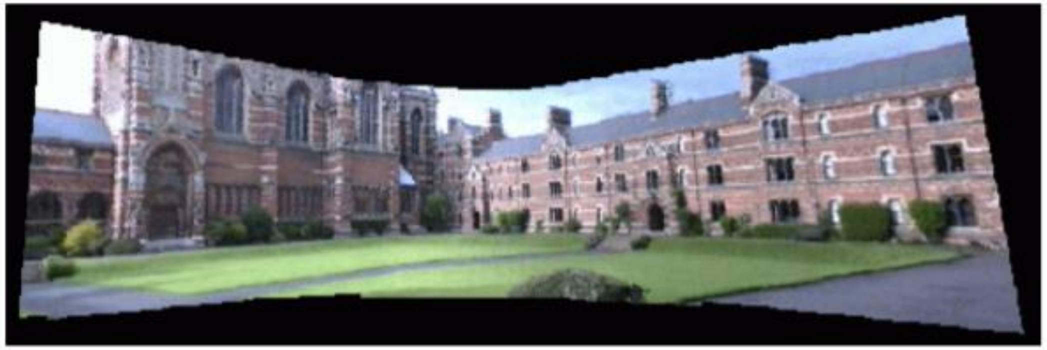


$$(x, y, 1)^T = x \propto PX = K[r_1 r_2 \cancel{r_3} t]\begin{pmatrix} X \\ Y \\ \cancel{0} \\ 1 \end{pmatrix} = H\begin{pmatrix} X \\ Y \\ 1 \end{pmatrix}$$

Rotating camera, arbitrary world







$$(x, y, 1)^T = x \propto PX = K(r_1 r_2 r_3 \cancel{t})\begin{pmatrix} X \\ Y \\ Z \\ \cancel{1} \end{pmatrix} \propto KRK^{-1}x' = Hx'$$

(*iso*=same, *metric*=measure)

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} \varepsilon\cos\theta & -\sin\theta & t_x \\ \varepsilon\sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} \qquad \varepsilon = \pm 1$$

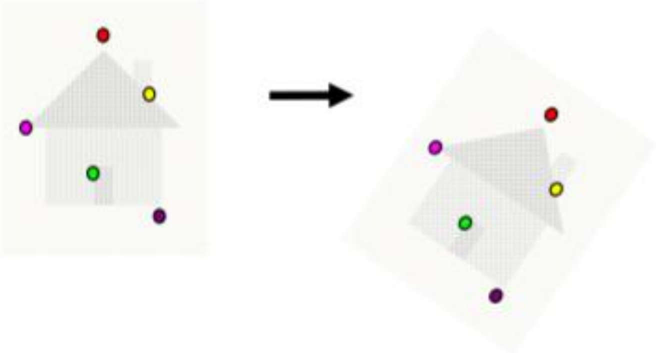
orientation preserving: $\varepsilon = 1$
orientation reversing: $\varepsilon = -1$

<u>isometries</u>

$$\mathrm{x}' = \mathbf{H}_E\,\mathrm{x} = \begin{bmatrix} \mathbf{R} & \mathrm{t} \\ 0^\mathsf{T} & 1 \end{bmatrix} \mathrm{x} \qquad \mathbf{R}^\mathsf{T}\mathbf{R} = \mathbf{I}$$

3DOF (1 rotation, 2 translation)

special cases: pure rotation, pure translation



Invariants: length, angle, area

Homography – 单应变换

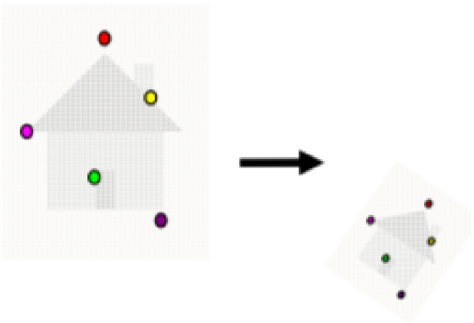
$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

(isometry + scale)

<u>similarities</u>

$$\mathrm{x}' = \mathbf{H}_S\, \mathrm{x} = \begin{bmatrix} s\mathbf{R} & \mathrm{t} \\ 0^{\mathsf{T}} & 1 \end{bmatrix} \mathrm{x}$$

$$\mathbf{R}^{\mathsf{T}}\mathbf{R} = \mathbf{I}$$



4DOF (1 scale, 1 rotation, 2 translation)

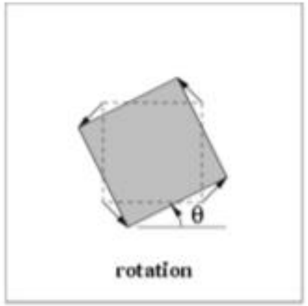
also know as *equi-form* (shape preserving)

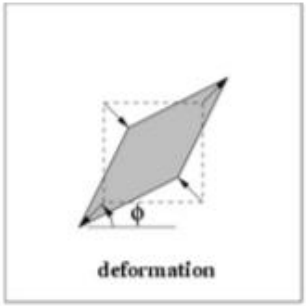
metric structure = structure up to similarity (in literature)

Invariants: ratios of length, angle, ratios of areas, parallel lines

$$\begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

$$\mathrm{x}' = \mathbf{H}_A \, \mathrm{x} = \begin{bmatrix} \mathbf{A} & \mathrm{t} \\ 0^{\mathrm{T}} & 1 \end{bmatrix} \mathrm{x}$$





affine transformations

$$\mathbf{A} = \mathbf{R}(\theta)\mathbf{R}(-\phi)\mathbf{D}\mathbf{R}(\phi) \qquad \mathbf{D} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

6DOF (2 scale, 2 rotation, 2 translation)

non-isotropic scaling! (2DOF: scale ratio and orientation)

Invariants: parallel lines, ratios of parallel lengths,
ratios of areas

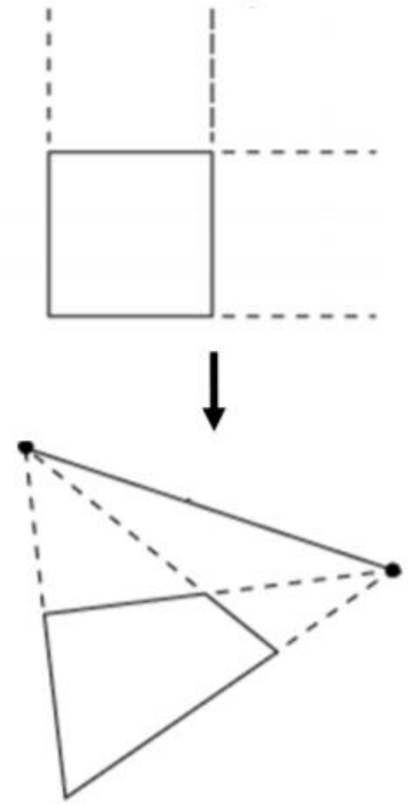
Homography – 单应变换

homographies

$$\mathbf{H}_P = \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} = \begin{pmatrix} \mathbf{A} & \vec{t} \\ \vec{v}^T & v \end{pmatrix}$$

Changes homogeneous coordinate!

$$\mathrm{x}' = \mathbf{H}_P \, \mathrm{x} = \begin{bmatrix} \mathbf{A} & \mathrm{t} \\ \mathrm{v}^\mathrm{T} & v \end{bmatrix} \mathrm{x} \qquad \mathrm{v} = (v_1, v_2)^\mathrm{T}$$



8DOF (2 scale, 2 rotation, 2 translation, 2 line at infinity)

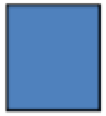
Acts non-homogeneous over the plane

Allows to observe vanishing points, horizon

Invariants: cross-ratio of four points on a line (ratio of ratio)

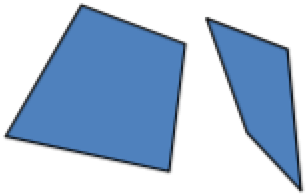
Homography – 单应变换

A square transforms to:



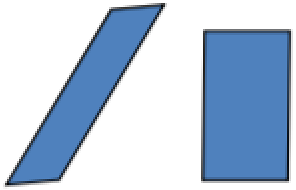
Projective
8dof

$$\begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix}$$



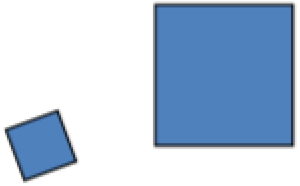
Affine
6dof

$$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



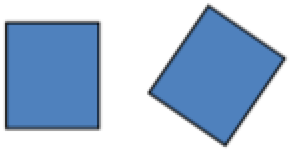
Similarity
4dof

$$\begin{bmatrix} sr_{11} & sr_{12} & t_x \\ sr_{21} & sr_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

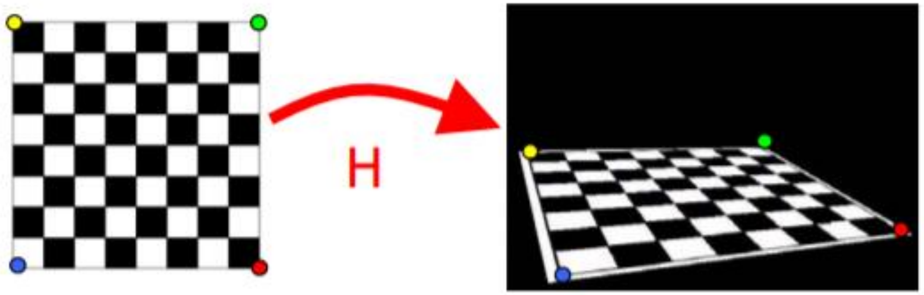


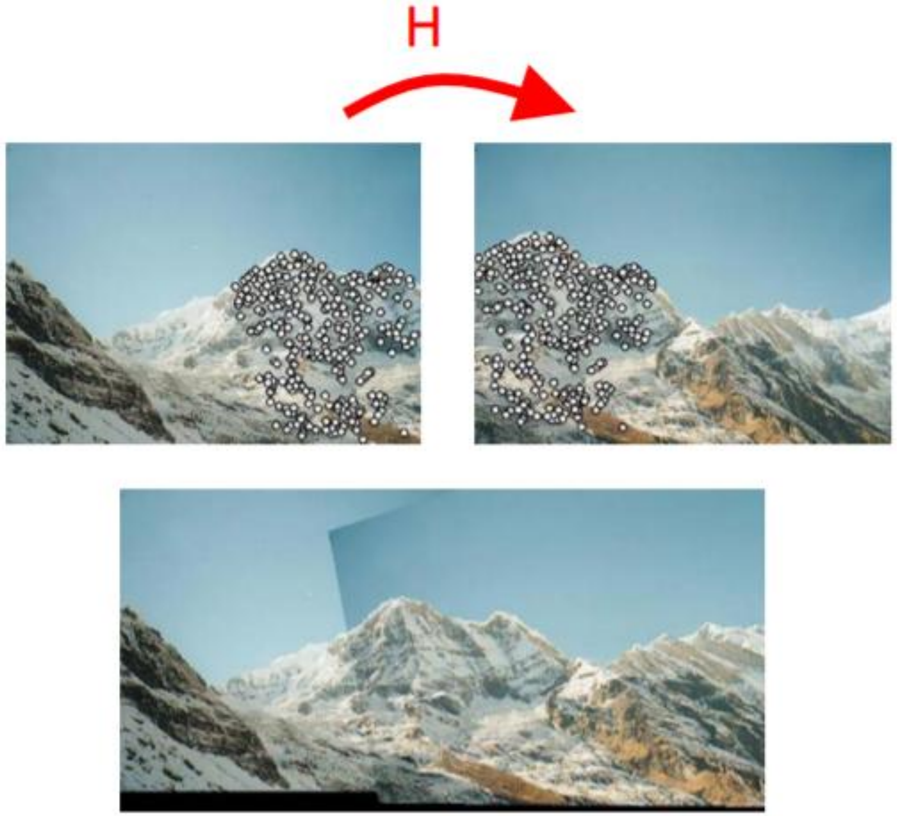
Euclidean
3dof

$$\begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$$



- Estimate homography from point correspondences between:
  - two images
  - model plane and image
- Assumption: planar motion!





- Homography mapping from image to image
  (holds under planar camera motion as mentioned before):

$$x' \propto Hx$$

$$\Leftrightarrow$$

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Homography – 单应变换

<u>Homography estimation</u>

- Equation for one point correspondence i:

$$\lambda \begin{bmatrix} x_i' \\ y_i' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

9 entries, 8 degrees of freedom
(scale is arbitrary)

i^{th} row of H

$$\lambda \mathbf{x}_i' = \mathbf{H}\,\mathbf{x}_i = \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \\ \mathbf{h}_3^T \end{bmatrix} \mathbf{x}_i$$

Homogeneous coordinate, might be 1

$$\mathbf{x}_i' \times \mathbf{H}\,\mathbf{x}_i = 0$$

$$\mathrm{x}_i' \times \mathrm{H}\mathrm{x}_i = \begin{pmatrix} y_i'\mathrm{h}^{3^\mathsf{T}}\mathrm{x}_i - w_i'\mathrm{h}^{2^\mathsf{T}}\mathrm{x}_i \\ w_i'\mathrm{h}^{1^\mathsf{T}}\mathrm{x}_i - x_i'\mathrm{h}^{3^\mathsf{T}}\mathrm{x}_i \\ x_i'\mathrm{h}^{2^\mathsf{T}}\mathrm{x}_i - y_i'\mathrm{h}^{1^\mathsf{T}}\mathrm{x}_i \end{pmatrix}$$

$$\begin{bmatrix} 0^\mathsf{T} & -w_i'\mathrm{x}_i^\mathsf{T} & y_i'\mathrm{x}_i^\mathsf{T} \\ w_i'\mathrm{x}_i^\mathsf{T} & 0^\mathsf{T} & -x_i'\mathrm{x}_i^\mathsf{T} \\ \cancel{-y_i'\mathrm{x}_i^\mathsf{T}} & \cancel{x_i'\mathrm{x}_i^\mathsf{T}} & \cancel{0^\mathsf{T}} \end{bmatrix} \begin{pmatrix} \mathrm{h}^1 \\ \mathrm{h}^2 \\ \cancel{\mathrm{h}^3} \end{pmatrix} = 0$$

3 equations, only 2 linearly independent, drop third row

Direct linear transform

$$\begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix} h = 0 \Rightarrow Ah = 0$$

- H has 8 DOF (9 parameters, but scale is arbitrary)
- One correspondence gives two linearly independent equations
- Four matches needed for a minimal solution (null space of 8x9 matrix)
- More points: search for “best” according to some cost function

- No exact solution because of inexact measurements due to noise
- With n correspondences: size A is 2nx9, rank most likely not 8

$$\begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_n \end{bmatrix} h = 0 \qquad Ah = 0$$

- Find approximate solution
  - Additional constraint needed to avoid 0, e.g. $\|h\| = 1$
  - $Ah = 0$ not possible, so minimize $\|Ah\|$

<u>Objective</u>

Given n≥4 2D to 2D point correspondences $\{x_i \leftrightarrow x_i'\}$, determine the 2D homography matrix H such that $x_i'=Hx_i$

<u>Algorithm</u>

(i) For each correspondence $x_i \leftrightarrow x_i'$ compute A_i. Usually only two first rows needed.

(ii) Assemble *n* 2x9 matrices A_i into a single 2*n*x9 matrix A

(iii) Obtain SVD of A. Solution for h is last column of V,
=singular value of A
=eigen vector to the smallest eigen value of A^TA

(iv) Determine H from h